ELARA: Reliability Gates Under Protocol Limits — A Cross-Benchmark Study of Drift-Signal Generalization in Multimodal Anomaly Fusion

Pratik Niroula
Independent Researcher

Draft thesis chapter, May 2026

Abstract

This chapter reports a scoped result for validation-derived drift gates in multimodal anomaly fusion under a locked audited-reanalysis policy. The same Kolmogorov–Smirnov drift signal that produces the audited mechanism endpoints on the label-aligned stress benchmark (ELARA-Bench-LA: +0.0506 zero-attack and +0.0319 max-attack all-domain deltas at locked τ=0.66, k-of-D sweep at k=4 mean-gate) does not produce a Holm-corrected significant clean-performance improvement across the inspected public-benchmark family. Under MVTec's canonical one-class protocol, supervised fusion baselines operate near chance because the train fold is normal-only; canonical PR-AUC / ECE / Brier values equal the canonical test-fold's anomaly prevalence rather than useful discrimination evidence (Phase 1.A audit verdict `METRICS_VALID_BUT_MISINTERPRETED`) and are omitted from promoted tables. Under the locked Phase 1 validation-frozen RGA+ head and validation-frozen primary comparator policy, MVTec 3D-AD PatchCore supervised-paired is non-significant (p_{Holm}=0.919), MVTec LOCO-AD PatchCore supervised-paired has a sign-flipped negative $\Delta = -0.008$ (not significant), VisA RGB+edge supervised-paired is non-significant (p_{Holm}=0.496), and UNSW-NB15 flow/conn/context retains Holm-corrected significance at a practically very small $\Delta = +0.0003$. The audited reanalysis therefore reports *mixed* outcomes on the inspected public-benchmark family; the dominant supported contribution is the coherent-collapse mechanism evidence on the label-aligned benchmark. None of these is independent confirmatory replication; confirmation requires Family D future locked evaluation on previously uninspected partitions.

The work is organized around ELARA (Evidence-Layered Anomaly Reliability Architecture), a research prototype for score-level multimodal anomaly fusion. The mechanism under study is RGA (Reliability-Gated Attention), a reliability-aware attention module that keeps the standard attention path under normal conditions and injects reliability weights when batch-level evidence suggests that one or more domains have degraded. This is a graduate thesis submission rather than as a marketing description of the system. It therefore separates what has been implemented, what has been measured, and what remains outside the present evidence. The primary benchmark is co-observed MVTec 3D-AD across all eight extracted categories (3,226 paired RGB/depth samples, no label alignment). The corrected MVTec run follows the canonical one-class protocol, and the held-out-category and M3DM-style variants are treated as diagnostics rather than new superiority claims. The secondary benchmark, ELARA-Bench-LA, combines four real-domain datasets via explicit label alignment; on it the same gate improves all-domain coherent-collapse attacks by +0.0506 and +0.0319 but the clean split is near saturated. A complete k-of-D corruption benchmark shows that the current mean gate has weak or mixed effects for isolated

and partial domain failures and activates fully only at all-domain collapse; the implemented minimum/hybrid RGA-v2 candidates do not yet close that gap at $\tau_{\min} = 0.34$. Public PatchCore supervised-paired and held-out MVTec variants improve the paired fusion task; base RGA still does not become the clean-score leader, while RGA+ gives bounded ROC-AUC gains.

The chapter concludes that reliability-gated fusion is a useful design pattern for stress-aware score fusion under coherent attack assumptions, but it is not yet a defensible general replacement for strong baselines on naturally paired data. The repository now includes engineering prerequisites for the next stage: incident-protocol validation, category-aware reliability references, deployment-time calibration monitoring, and a finite-sample switching certificate. The remaining research work is empirical: stronger RGB–3D and clinical domain experts, additional naturally co-observed anomaly datasets, and prospective deployment monitoring.

Contents

1 Introduction and Motivation

Operational anomaly detection is rarely a single-model problem. A fraud case may appear in transaction features, device behavior, account history, text evidence, or network activity. An industrial defect may appear in both color information and geometry. A security event may be weak in one log source and clearer in another. In these settings, the system designer must decide not only which domain models to train, but also how to combine their outputs once those models are deployed.

This chapter focuses on the second problem: score-level fusion under imperfect domain reliability. The assumption is that domain experts already produce scores, confidence values, and compact evidence vectors. The open question is how the fusion layer should behave when domain quality changes at inference time. A static fusion model can learn useful interactions from training data, but it may continue to trust a domain after that domain has shifted, collapsed, or become poorly calibrated. A simple confidence-weighted average is easy to understand, but it uses local confidence values without asking whether the domain itself still resembles the validation distribution.

ELARA was built to study this problem in a reproducible way. It is not presented as a new fraud detector, cybersecurity detector, text detector, or industrial inspection detector. Instead, it is a research system for combining heterogeneous anomaly evidence. The central component is RGA, a reliability-aware fusion module that uses three forms of evidence: validation calibration, test-time score-distribution drift, and score sharpness. The module is deliberately conservative. When domains appear reliable, the system uses the static attention model. When the average reliability drops below a threshold, the reliability-aware path is activated.

The chapter is in this form because the current evidence is both useful and bounded. The real-domain fusion benchmark combines fraud, cyber, behavior, and text datasets, but those records are aligned by label rather than by shared entities, timestamps, or incidents. It is useful for score-fusion stress testing, but it is not a complete multimodal incident benchmark. To reduce this gap, the repository now includes an all-category MVTec 3D-AD RGB/3D benchmark path. That path gives natural co-observation, while its canonical one-class protocol and lightweight score construction make it a protocol diagnostic rather than a final performance claim.

1.1 Research Question

The research question is:

> *Can a multimodal anomaly-fusion system preserve competitive clean score-fusion performance while using calibration and drift evidence to reduce brittle behavior under domain reliability stress?*

This question is intentionally narrower than broad anomaly-detection superiority. The chapter does not claim that RGA is always better than static attention, that the current paired MVTec run is complete, or that the synthetically paired benchmark represents naturally co-observed incidents. The claim is about the fusion layer under specific stress conditions.

1.2 Main Claim

The main thesis of this chapter is:

RGA is most useful as a stress-response mechanism: it preserves the static attention path when reliability evidence is high, and it helps under coherent score-collapse conditions when reliability evidence indicates domain failure.

The measured evidence supports this claim on the secondary benchmark. The clean table is near saturated and should be treated as a sanity check. The reliability-stress experiments, the threshold sweep, and the component ablation provide the more useful evidence about the mechanism.

1.3 Contributions

We make four scoped contributions:

1. It describes ELARA, a score-level anomaly-fusion research system with a unified long-format schema for heterogeneous domain evidence.
2. It defines RGA, a conservative reliability-gated attention module that combines masked attention with calibration, drift, and sharpness signals.
3. It evaluates the method against practical score-fusion baselines, including random forest fusion, early-fusion MLP, late-fusion ensemble, confidence-weighted mean, score-level Tent, and pseudo-label TTT.
4. It reports an all-category paired MVTec 3D-AD benchmark, multi-seed confidence intervals, mechanism ablations, failure cases, and explicit threats to validity.

1.4 Evidence Boundary

The evidence boundary is part of the contribution because it prevents the results from being overstated. The real-domain benchmark uses real local datasets, but the records are label-aligned. The MVTec benchmark uses natural RGB/3D pairing across eight categories, but it still uses lightweight normal-reference image statistics and the canonical one-class anomaly protocol. These two artifacts answer different questions. The first is useful for reliability-stress score fusion. The second is useful for exposing protocol limits on naturally paired data. Neither should be treated as a complete final benchmark for all multimodal anomaly detection.

2 Background and Related Work

2.1 Anomaly Detection as a Systems Problem

Deep anomaly detection has been studied across tabular, sequential, image, text, and graph data. A common difficulty is that the definition of an anomaly depends on the application, the label source, the imbalance level, and the cost of false positives and false negatives. Pang et al. [3] emphasize that anomaly detection methods cannot be evaluated without attention to context. This chapter follows that view. The purpose is not to replace domain-specific detectors. The purpose is to study what happens after those detectors produce evidence that must be fused.

2.2 Multimodal and Score-Level Fusion

Multimodal learning is often described in terms of early fusion, late fusion, hybrid fusion, and interaction-based fusion [2]. Early fusion joins features before prediction. Late fusion combines decisions or scores from separate models. Stacking trains a meta-model over domain outputs. Attention-based fusion gives the model a way to learn interactions among domains, especially when some domains are missing. The weakness is that standard attention learns a trust policy from the training distribution and does not automatically know when a domain has become unreliable at inference time.

The score-level setting used in this chapter is intentionally practical. Many deployed systems cannot easily retrain all domain experts together. Different teams may own different detectors, and those detectors may expose only scores, calibrated probabilities, or compact embeddings. A score-level fusion study is therefore narrower than raw multimodal learning, but it matches a common engineering constraint.

2.3 Calibration, Drift, and Reliability

Calibration asks whether predicted probabilities match empirical frequencies. Post-hoc calibration methods such as Platt scaling, isotonic calibration, and modern neural calibration analyses show that probability quality is separate from ranking quality [4, 5, 6]. Uncertainty and calibration can degrade under dataset shift [7]. For a fusion system, this matters because a domain can remain numerically confident while becoming less trustworthy.

RGA treats calibration as one signal, not as the whole reliability definition. The reliability estimator also compares current score distributions against validation score distributions using a Kolmogorov-Smirnov statistic and includes a score-sharpness term. This design is not meant to be a complete theory of reliability. It is a simple, measurable rule that can be tested under controlled stress.

2.4 Test-Time Adaptation Baselines

Test-time adaptation updates a model during inference to handle distribution shift without access to labels at test time. Two threads of this literature matter for this work, and both are represented in the baseline suite.

Test-time training (TTT). Sun et al. [8] proposed updating a model on each test sample by minimizing a self-supervised auxiliary loss (the original work used rotation prediction). The intuition: an auxiliary task trained jointly with the main loss anchors the representation, so adapting the auxiliary component at test time also adjusts the main predictor in a useful direction. Later variants relax the self-supervision to pseudo-labels: the model produces a confident prediction on the test sample, treats it as the label, and updates the parameters with a small step against that target. The pseudo-label adapter in this study (`ttt_pseudo_label_adapter`) is a score-level version of this approach: it acts on the fusion-head logits rather than on a deep representation.

Entropy-minimization TTA. Wang et al. [9] introduced Tent, which adapts only the batch normalization affine parameters of a frozen model to minimize the entropy of test predictions. Tent is simpler than full TTT and is the cleanest articulation of the hypothesis that confident predictions are correct predictions — formally, that the local entropy of the predictive distribution is a sufficient signal for which direction to update. The Tent score adapter in this study (`tent_score_adapter`) implements the score-level analog: a frozen fusion head is adapted by a scalar temperature and bias fitted to minimize per-sample entropy on the test batch.

EATA and SAR. The score-level baseline suite also includes EATA [22], which adds reliable-sample selection and non-redundancy filtering on top of Tent, and SAR [23], which adds per-sample entropy weighting with sharpness-aware minimisation on top of Tent. Both are implemented as score-level adapters operating on the static fusion head's logits; they appear in every result table that lists the test-time-adaptation comparator family.

Position of RGA relative to TTA. Both Tent and TTT adapt the *prediction* at test time — they ask which output should be revised. RGA sits one level up: it adapts the *weighting* of upstream domain scorers, not the fusion head itself. The reliability signals (validation ECE, KS drift, sharpness) are properties of the upstream domains' behavior, and the gate decides whether to trust them collectively or to fall back to a static weighting learned from training data. This makes RGA conceptually closer to selective prediction [17] — a controller that decides when to commit to the model's standard behavior — than to TTA in the Tent sense. Empirically the chapter compares all three (Tent, TTT, RGA) on both benchmarks so the contrast is visible.

2.5 RGB–3D Industrial Anomaly Detection

MVTec 3D-AD provides naturally paired RGB and 3D observations for industrial anomaly detection and localization [11]. Recent work on RGB–3D anomaly detection has shown that paired color and geometry can be useful when the detector is designed for that modality pair. M3DM uses hybrid feature and decision fusion with memory banks [12]. Crossmodal feature mapping uses agreement between RGB and point-cloud features as an anomaly cue [13]. Real3D-AD expands point-cloud anomaly benchmarking [14], and M3DM-NR studies noisy-resistant multimodal anomaly detection [15].

The MVTec path in this repository is not a replacement for those specialized RGB–3D detectors. It is a paired-data input path for the score-fusion layer. This distinction is important. The current MVTec smoke run asks whether the repository can construct and evaluate paired score-fusion inputs. It does not claim that the lightweight scorers compete with specialized RGB–3D methods.

3 System and Method

3.1 System Boundary

ELARA is organized around the idea that each sample may have several domain observations. A domain can be fraud, cyber telemetry, behavior, text, RGB imagery, depth/XYZ geometry, or another source that can produce a score and compact evidence. The fusion layer receives these domain rows in a common schema. It then produces one anomaly probability and optional domain-level attributions.

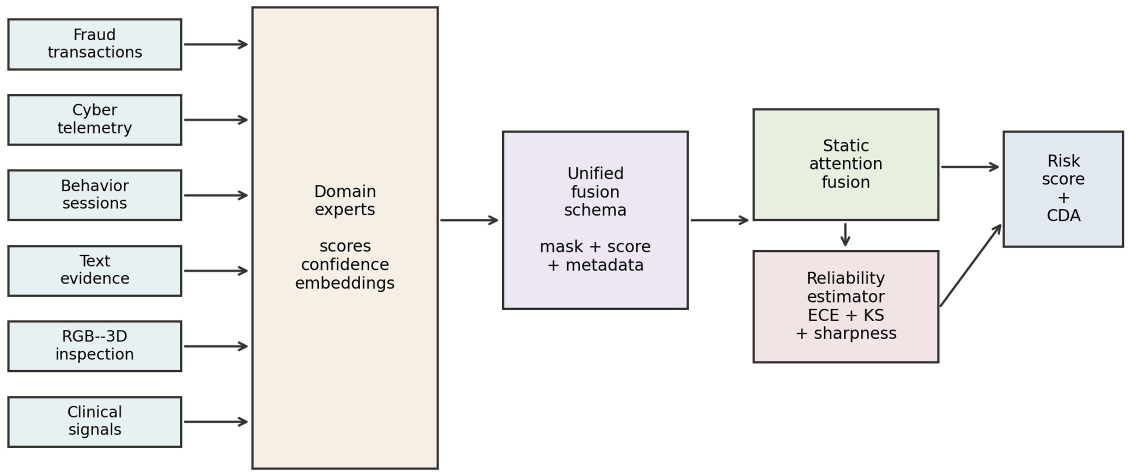

Figure 1: ELARA system architecture. Domain experts produce scores, confidence values, and compact embeddings. The fusion schema feeds static attention and the reliability estimator. The output is a risk score plus counterfactual domain attribution.

The system design separates domain learning from fusion learning. This is a deliberate simplification. It lets the research focus on reliability-aware fusion without requiring every raw domain model to be retrained as part of one large multimodal network.

3.2 Input Schema

The long-format fusion schema contains one row per sample-domain pair. Each row contains a sample identifier, a domain name, a binary label when available, a score, a confidence value, and embedding columns. Missing domains are represented by masks, not by fabricated raw observations. This allows a sample with all domains, a sample with one missing domain, and a sample with only one available domain to share the same model interface.

For the label-aligned real-domain benchmark, the metadata records the field `natural_pairing` as `false`. For the MVTec 3D-AD path, the same field is `true` because the RGB and 3D observations come from the same object scan. This metadata distinction is important because the same fusion code can be used for two different kinds of evidence.

3.3 Problem Formulation

Let sample i contain up to D domain observations. Domain d contributes a feature vector $x_{i,d} \in \mathbb{R}^F$ and a missing-domain mask $m_{i,d}$. The system estimates

$$\hat{p}_i = P(y_i = 1 \mid \{x_{i,d}, m_{i,d}\}_{d=1}^{D}), \tag{1}$$

where $\hat{p}_i$ is the final anomaly probability. A static fusion model learns this relationship from the training data and uses the same learned trust policy at inference time. RGA adds a batch-level reliability vector $r \in [0,1]^D$ in the reported runs and applies it through each sample's observed-domain mask. A per-sample reliability mode is implemented for ablation, but the tables in this chapter use the batch-level gate.

3.4 Static Attention Path

The static attention path encodes each domain vector into a shared latent space. A learned domain embedding allows the model to distinguish evidence sources. A multi-head self-attention block [1] then operates across available domains while ignoring missing domains through the mask. This path is the default because reliability adaptation should not disturb predictions when the domains look normal.

3.5 Reliability Estimation

The reliability estimator is fit after model training. For each domain it stores validation score distributions and calibration information. At inference time, the reliability for domain d is

$$r_d = \alpha(1 - \mathrm{ECE}_d) + \beta p_{\mathrm{KS},d} + \gamma S_d, \tag{2}$$

where ECE_d is validation expected calibration error, $p_{\mathrm{KS},d}$ is the Kolmogorov-Smirnov score-drift p-value, and S_d is score sharpness based on squared distance from 0.5. The current configuration uses $(\alpha, \beta, \gamma) = (0.45, 0.35, 0.20)$.

This formula is simple by design. It is not a final answer to reliability estimation. It is a testable rule that separates three observable signals: how well a domain calibrated on validation data, whether the current batch resembles the validation score distribution, and whether scores are informative or close to the uninformative midpoint.

3.6 Reliability Gate

The gate is the main difference between RGA and always-on reliability weighting. Let $\bar{r}$ be the mean reliability over present domains and let $\tau = 0.66$. If $\bar{r} \geq \tau$, the system uses the static attention path. If $\bar{r} < \tau$, reliability weights are injected into the fusion block:

$$\tilde{r}_{i,d} = (1 - m_{i,d}) r_d. \tag{3}$$

Equivalently, if unreliability is written as $u = 1 - \bar{r}$, the same rule is $u > 1 - \tau = 0.34$. The unreliability threshold is therefore not 0.66; using the same numeric threshold for both $\bar{r}$ and u would reverse the gate logic.

This design is conservative. It assumes that static attention is already a strong clean model and that reliability weights should be used only when there is evidence of degradation. This also makes the method easier to criticize and test: if the gate rarely activates, the method should behave like static attention; if the gate activates under stress, the stress outcomes should change.

The codebase also now exposes an RGA-v2 gate family for follow-up experiments: a minimum-reliability gate $G_{\min} = ⊮[\min_d r_d < \tau_{\min}]$ and a hybrid gate $G_{\mathrm{hyb}} = ⊮[\bar{r} < \tau_{\mathrm{mean}} \text{ or } \min_d r_d < \tau_{\min}]$. These variants are diagnostic candidates in the current chapter, not validated fixes. The completed k-of-D run shows that with $\tau_{\min} = 0.34$, observed minimum reliability stays above the minimum threshold, so the minimum branch never activates and the hybrid gate behaves like the original mean gate. The reported base-RGA tables keep the original mean gate unless a configuration explicitly selects `minimum` or `hybrid`.

3.7 Switching-Rule Certificate and Scope

The deployment question is whether the gated policy is preferable to static attention on the cases where the gate actually fires. Let $\ell_s(i)$ be the loss of the static path, $\ell_r(i)$ the loss of the reliability-aware path, and $g_i \in \{0,1\}$ the gate decision. The realized gated loss is

$$\ell_g(i) = (1 - g_i)\ell_s(i) + g_i\ell_r(i). \tag{4}$$

On an audited validation window, the switch is certified only when the fired subset has a positive lower confidence bound on paired benefit. For $X_i = \ell_s(i) - \ell_r(i)$ and $\mathcal{F} = \{i : g_i = 1\}$, the empirical benefit is

$$\hat{\mu}_{\mathcal{F}} = \frac{1}{|\mathcal{F}|}\sum_{i\in\mathcal{F}} X_i. \tag{5}$$

The switch is certified only if

$$\mathrm{LCB}_\alpha = \hat{\mu}_{\mathcal{F}} - \mathrm{margin}_\alpha > 0. \tag{6}$$

The implementation is `bounded_switching_certificate`. This is a finite-sample deployment audit, not a universal population-level proof. The population theorem in Appendix C is instead stated as a risk-dominance condition with explicit detection power, false-fire rate, benefit, and clean-operation cost terms.

3.8 Counterfactual Domain Attribution

The explanation path masks one available domain at a time and recomputes the prediction. For sample i, let $\hat{p}_i$ be the original prediction and $\hat{p}_{i,-d}$ be the prediction when domain d is masked. The attribution is

$$\Delta_{i,d} = \hat{p}_i - \hat{p}_{i,-d}. \tag{7}$$

A positive value means that the domain increased anomaly risk. A negative value means that the domain reduced anomaly risk. This is not a feature-level explanation; it is a domain-level explanation for analysts who need to know which source of evidence mattered most.

4 Benchmark Construction

4.1 Naturally Paired MVTec 3D-AD Benchmark

The paired benchmark path uses MVTec 3D-AD because each object observation has paired RGB and 3D/XYZ evidence. The preparation script discovers paired files, emits two rows per observation, stores the original pairing key, and marks the metadata as naturally paired. The current run covers all eight extracted object categories – bagel, cable_gland, cookie, dowel, foam, peach, rope, and tire – for a total of 3,226 paired samples and 6,452 domain rows, with a positive fraction of 0.224 and full RGB and depth coverage.

Table 1: Naturally paired MVTec 3D-AD multi-category metadata.

Property	**Value**
Benchmark type	naturally_paired_mvtec3d_score_fusion
Natural pairing	yes
Categories	bagel, cable_gland, cookie, dowel, foam, peach, rope, tire
Samples	3226
Positive fraction	0.224
Score fit split	train/good
Score fit samples	2070
Score normalization	train_good_distance_minmax_clipped
Domains	rgb, depth_or_xyz

The initial MVTec domain scores are lightweight normal-reference image-statistic scores fit from original train-good observations only. This keeps the benchmark locally runnable, but it limits the absolute performance ceiling. The corrected attention-fusion split follows the canonical MVTec one-class protocol: train/validation rows are normal-only and the test fold is mixed. The result in Table 2 is therefore a protocol diagnostic rather than a conventional supervised leaderboard. Random forest, late fusion, Tent, TTT, static attention, and RGA have little two-class training signal; the small RGA delta should not be read as a general win. The held-out-category and M3DM-style variants probe the same limitation under different paired-data conditions. The updated held-out-category protocol now reserves positive validation rows from train categories, which lets RGA+ tune without seeing the held-out categories; it improves ROC-AUC slightly but does not settle PR-AUC or category transfer. These variants remain part of the evidence boundary discussed in §6.3.

Table 2: Naturally paired MVTec 3D-AD clean held-out performance across eight categories. Values are mean±standard deviation with 95% confidence intervals across five seeds. Under the canonical one-class protocol, this table is a supervised-fusion protocol diagnostic rather than a two-class leaderboard.

Method	**ROC-AUC mean**	**ROC-AUC 95% CI**
Random forest	0.5000	[0.5000, 0.5000]
Conf.-weighted mean	0.4458	[0.4458, 0.4458]
Tent score adapter	0.5000	[0.5000, 0.5000]
TTT pseudo-label	0.5000	[0.5000, 0.5000]
Early fusion MLP	0.5455	[0.5180, 0.5836]
Late fusion ensemble	0.5000	[0.5000, 0.5000]
Static attention	0.5424	[0.4992, 0.5661]
RGA attention	0.5609	[0.5394, 0.5842]
RGA+ boosted fusion	0.5000	[0.5000, 0.5000]
RGA+ router	0.5609	[0.5394, 0.5842]

4.2 Secondary Label-Aligned Real-Domain Benchmark

The secondary benchmark, ELARA-Bench-LA, combines four real local domains: credit card fraud, cyber telemetry, online behavior, and labeled news text. Domain scorers are trained first, and their out-of-fold scores are converted into the common fusion schema. The dataset contains 8,000 composite samples, four domains, eight embedding features per domain, a 0.307 positive rate, and 11.8% nominal missingness. Domain coverage is approximately balanced across the four domains.

Table 3: Benchmark metadata for the secondary synthetically paired fusion input.

Property	Value
Benchmark type	label_aligned_real_domain_score_fusion
Natural pairing	no
Pairing unit	binary-label-aligned composite sample
Samples	8000
Positive fraction	0.301
Scorer train fraction	0.05
Source-row disjoint splits	yes
Split column	fusion_split
Domains	fraud, cyber, behavior, nlp

Table 4: Out-of-fold domain scorer quality used to construct ELARA-Bench-LA.

Domain	Rows	Pos. rate	OOF ROC	OOF PR
fraud	20000	0.025	0.9126	0.6961
cyber	20000	0.500	0.9539	0.9546
behavior	12330	0.155	0.7978	0.4389
nlp	20000	0.500	0.9926	0.9928

This benchmark is useful because it uses real domain datasets and lets the fusion layer be tested under controlled stress. It is also limited because the domains are the secondary benchmark rather than naturally co-observed. A positive composite draws positive records from available domains, and a negative composite draws negative records. This creates a meaningful score-fusion stress test, but it does not teach the model real temporal or causal relationships among the domains. Source rows are now split into train, validation, and test pools before composite sampling, and the experiment runner consumes the `fusion_split` column to keep the same `domain/source_row/label` key out of multiple supervised fusion splits.

4.3 UNSW-NB15 Naturally Structured Event-View Benchmark

UNSW-NB15 provides an additional naturally structured event-view evaluation outside visual paired data. Each row is a single network event with three software-derived view modalities:

flow timing / throughput / inter-packet statistics (13 features)

connection protocol and session structure (14 features)

context aggregated past-window connection counters (12 features)

The three views are computed from the same event identifier; their pairing strength under the Phase-0.6 classification is `naturally_structured_views` rather than `independent_modalities`. Splits are stratified random draws on (attack category, label) using event id; the test fold is disjoint from train and validation.

Table 5: UNSW-NB15 five-seed clean ROC-AUC over three software-derived views (flow / connection / context) of the same network event, 55,491 paired events. Reported descriptively; method ordering follows the clean-table convention.

Method	ROC-AUC	PR-AUC	F1	ECE	Brier
Random forest	0.9890±0.0001 [0.9889, 0.9891]	0.9939 [0.9939, 0.9940]	0.9522±0.0002 [0.9520, 0.9524]	0.0172±0.0003 [0.0169, 0.0176]	0.0420±0.0001 [0.0419, 0.0422]
Conf.-weighted mean	0.3486 [0.3486, 0.3486]	0.5454 [0.5454, 0.5454]	0.7798 [0.7798, 0.7798]	0.4070 [0.4070, 0.4070]	0.4293 [0.4293, 0.4293]
Tent score adapter	0.9543 [0.9543, 0.9543]	0.9722 [0.9722, 0.9722]	0.9133 [0.9133, 0.9133]	0.0519 [0.0519, 0.0519]	0.0854 [0.0854, 0.0854]
TTT pseudo-label	0.9482 [0.9482, 0.9482]	0.9687 [0.9687, 0.9687]	0.9131 [0.9131, 0.9131]	0.0417 [0.0417, 0.0417]	0.0873 [0.0873, 0.0873]
Early fusion MLP	0.9855±0.0002 [0.9851, 0.9858]	0.9918±0.0001 [0.9916, 0.9920]	0.9451±0.0005 [0.9447, 0.9459]	0.0258±0.0031 [0.0218, 0.0300]	0.0489±0.0006 [0.0481, 0.0497]
Late fusion ensemble	0.9344 [0.9344, 0.9344]	0.9592 [0.9592, 0.9592]	0.9109 [0.9109, 0.9109]	0.0780 [0.0780, 0.0780]	0.1027 [0.1027, 0.1027]
Static attention	0.9790±0.0007 [0.9783, 0.9800]	0.9879±0.0004 [0.9876, 0.9885]	0.9358±0.0016 [0.9340, 0.9382]	0.0145±0.0107 [0.0076, 0.0334]	0.0554±0.0015 [0.0533, 0.0576]
RGA attention	0.9748±0.0036 [0.9690, 0.9786]	0.9851±0.0024 [0.9812, 0.9876]	0.9333±0.0025 [0.9294, 0.9367]	0.0107±0.0042 [0.0063, 0.0176]	0.0589±0.0035 [0.0553, 0.0648]
RGA+ boosted fusion	0.9891±0.0002 [0.9890, 0.9894]	**0.9940±0.0001 [0.9939, 0.9941]**	0.9510±0.0011 [0.9494, 0.9525]	**0.0060±0.0005 [0.0055, 0.0067]**	**0.0404±0.0004 [0.0400, 0.0409]**
RGA+ router	**0.9893±0.0003 [0.9889, 0.9896]**	**0.9941±0.0002 [0.9937, 0.9943]**	**0.9526±0.0006 [0.9517, 0.9533]**	0.0140±0.0066 [0.0062, 0.0243]	0.0412±0.0015 [0.0400, 0.0436]

In the locked audited reanalysis, validation-frozen RGA+ (router) obtains the highest point estimate against the validation-frozen primary comparator (random forest), with a practically very small ROC-AUC delta of approximately +0.0003. The fixed-representative-seed DeLong test is statistically distinguishable after Family A audited-primary $K{=}5$ Holm correction, but this is not independent confirmatory replication and does not establish broad cross-domain superiority. Confirmation requires a Family D future locked evaluation on a previously uninspected partition.

4.4 Baselines

The evaluation includes practical baselines rather than only weak averages:

- confidence-weighted mean fusion, which directly weights scores by local sharpness;
- early-fusion MLP, which learns from flattened domain features and missing indicators;
- late-fusion ensemble, which stacks per-domain predictions;
- random forest fusion [16], a strong structured-data baseline;
- static attention, the same attention architecture without reliability adaptation;
- Tent score adapter, a score-level entropy-minimization baseline;
- TTT pseudo-label adapter, a score-level high-confidence adaptation baseline.

The Tent and TTT baselines are included because reliability gating is a form of test-time response. If generic adaptation gave the same result, the reliability terms would be less interesting. In the current clean table, these baselines are competitive, which is expected on the saturated split. The stress tests are more important for distinguishing the mechanisms.

The post-audit hard-mode RealFusion-LA run uses `-scorer-train-fraction 0.05` and is archived at `experiments/fusion/craf_real_results_hard.json`. It weakens the fraud and behavior domain scorers (OOF ROC-AUC 0.913 and 0.798), but the label-aligned fusion task remains near saturated: static attention and RGA both reach mean ROC-AUC 0.99896 across five seeds, and RGA improves drift degradation area on zero domains. This is evidence about the benchmark limitation, not a new superiority claim.

4.5 Metrics and Statistical Reporting

The chapter reports ROC-AUC, PR-AUC, F1, expected calibration error, Brier score, and accuracy where available. ROC-AUC is useful for ranking. PR-AUC is important because anomaly tasks can be imbalanced [10]. F1 summarizes thresholded classification. ECE and Brier score provide calibration information.

Clean results are summarized across seeds 42 through 46. Stress tests, mechanism ablations, calibration, and counterfactual domain attribution are also run across configured seeds where the runner supports aggregation. Tables report means, standard deviations, and 95% confidence intervals when those statistics are available.

4.6 Locked Audited-Reanalysis Policy and Future Replication Boundary

All current public-benchmark result cells in this chapter were inspected before the Phase 0.6 policy lock (`docs/research/audit/PHASE_0_6_FINAL_POLICY_LOCK.md`). Under that lock:

- **Current corrected results are locked audited reanalysis only**, not independent confirmatory replication.
- The RGA+ headline head is selected by validation ROC-AUC and frozen before test evaluation (router or boosted-fusion head; tie-break boost). Selection of the head is never read from the test fold.
- The primary comparator per cell is selected by validation ROC-AUC and frozen before test evaluation. Selection of the comparator is never read from the test fold; no "best-non-router" framing chosen from test winners is admitted.
- Analysis families:
  - **Family A**: audited public-benchmark reanalyses (MVTec 3D-AD, MVTec LOCO-AD, VisA RGB+edge proxy, UNSW-NB15). K=5 audited-primary cells receive a Holm-Bonferroni-corrected fixed-representative-seed DeLong p-value. Family A is never called *confirmatory*.
  - **Family B**: mechanism evidence on ELARA-Bench-LA at locked τ=0.66; the audited mechanism endpoints (B1 zero-attack all-domain, B2 max-attack all-domain) are reported from the k-of-D sweep at k=4 mean-gate (+0.0506 and +0.0319 respectively; `experiments/fusion/craf_real_k_domain_results.json`, PRIMARY per `PHASE_1_1_PRIMARY_RUN_RESOLUTION.md`). The standard-protocol adversarial table is a descriptive subsidiary surface only.
  - **Family C**: exploratory audits (Real3D-AD, noise-floor variants, held-out attack categories, local healthcare replay). Descriptive only; no superiority claim.
  - **Family D**: future unseen *confirmatory replication* only. Family D is the only family that may be described as pre-registered or confirmatory. Family D evaluations have not yet been executed; their permitted candidates are listed in `docs/research/audit/EXPERIMENT_REGISTRY.csv`.
- Current Family A inferential summaries use a *fixed-representative-seed* (seed 42) audited DeLong p-value because raw per-seed test prediction vectors are not archived for legacy result JSONs. Phase 2 must archive prediction arrays per seed and conduct paired sample-bootstrap inference

on a newly locked replication; until then the current p-values are bounded audited summaries, not confirmatory evidence.

- No universal ELARA claim is made: in particular, the chapter does not assert leaderboard-leadership, production-grade deployment readiness, clinical-deployment validation, or broad cross-domain superiority. The current evidence boundary supports the mechanism finding on the label-aligned stress benchmark and the mixed audited outcomes on the inspected public-benchmark family; nothing further.

The polarity probe is a validation-only score-orientation diagnostic; per the Phase 1.F lock it is logged but does not alter primary reported predictions (`docs/research/audit/POLARITY_DIAGNOSTIC_REPORT.md`). Per-domain reliability perturbation is reported as *model-response sensitivity* (input-sensitivity analysis), not as a causal effect on real anomaly incidence or operational safety.

5 Experiments and Results

5.1 Clean Performance Is a Sanity Check

Table 6 gives clean held-out performance on ELARA-Bench-LA. This table is intentionally not the headline evidence. Most methods are near the top of the ROC-AUC and PR-AUC scale. Static attention and RGA are effectively tied. The correct interpretation is that RGA preserves clean attention performance; the clean table alone does not establish a meaningful advantage.

Table 6: Saturated clean held-out performance on ELARA-Bench-LA. Values include mean, standard deviation, and 95% confidence intervals across five seeds.

Method	**ROC-AUC**	**PR-AUC**	**F1**	**ECE**	**Brier**
Random forest	0.9991±0.0001 [0.9990, 0.9991]	**0.9986±0.0001 [0.9985, 0.9987]**	**0.9900±0.0011 [0.9887, 0.9916]**	0.0341±0.0007 [0.0331, 0.0347]	0.0087±0.0001 [0.0087, 0.0088]
Conf.-weighted mean	0.9910 [0.9910, 0.9910]	0.9769 [0.9769, 0.9769]	0.9216 [0.9216, 0.9216]	0.0830 [0.0830, 0.0830]	0.0462 [0.0462, 0.0462]
Tent score adapter	0.9988 [0.9988, 0.9988]	0.9978 [0.9978, 0.9978]	0.9803 [0.9803, 0.9803]	0.0102 [0.0102, 0.0102]	0.0106 [0.0106, 0.0106]
TTT pseudo-label	0.9987 [0.9987, 0.9987]	0.9976 [0.9976, 0.9976]	0.9812 [0.9812, 0.9812]	0.0076 [0.0076, 0.0076]	0.0091 [0.0091, 0.0091]
Early fusion MLP	0.9990±0.0002 [0.9987, 0.9991]	0.9979±0.0003 [0.9975, 0.9982]	0.9828±0.0022 [0.9797, 0.9854]	0.0125±0.0023 [0.0090, 0.0149]	0.0116±0.0012 [0.0097, 0.0129]
Late fusion ensemble	0.9987 [0.9987, 0.9987]	0.9979 [0.9979, 0.9979]	0.9855 [0.9855, 0.9855]	0.0138 [0.0138, 0.0138]	0.0091 [0.0091, 0.0091]
Static attention	0.9990±0.0001 [0.9988, 0.9990]	0.9981±0.0001 [0.9979, 0.9982]	0.9815±0.0017 [0.9795, 0.9842]	**0.0053±0.0008 [0.0041, 0.0062]**	0.0083±0.0004 [0.0077, 0.0088]
RGA attention	0.9990±0.0001 [0.9988, 0.9990]	0.9981±0.0001 [0.9979, 0.9982]	0.9815±0.0017 [0.9795, 0.9842]	**0.0053±0.0008 [0.0041, 0.0062]**	0.0083±0.0004 [0.0077, 0.0088]
RGA+ boosted fusion	**0.9992 [0.9992, 0.9992]**	0.9986 [0.9986, 0.9986]	0.9855 [0.9855, 0.9855]	0.0060 [0.0060, 0.0060]	0.0070 [0.0070, 0.0070]
RGA+ router	**0.9992 [0.9991, 0.9992]**	0.9986 [0.9986, 0.9986]	0.9859±0.0008 [0.9855, 0.9873]	0.0083±0.0045 [0.0060, 0.0162]	**0.0070±0.0002 [0.0067, 0.0070]**

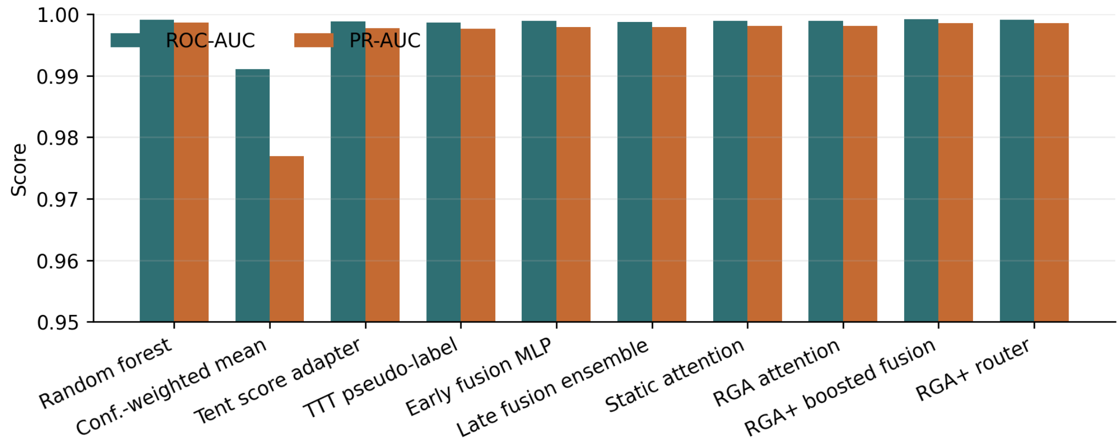

Figure 2: Clean benchmark ROC-AUC and PR-AUC across score-level and attention fusion methods. The compressed y-axis reflects the near-saturated clean split.

The clean results do show useful tradeoffs. Random forest fusion obtains the best F1 in the current artifact. Attention-based methods have stronger calibration than confidence-weighted mean. Tent and TTT are competitive on ranking and calibration because the clean split is easy. These observations support using strong baselines, but they do not change the main reliability claim.

Table 7: Clean ranking and F1 confidence intervals across configured seeds.

Method	**ROC-AUC 95% CI**	**PR-AUC 95% CI**	**F1 95% CI**
Random forest	[0.9990, 0.9991]	[0.9985, 0.9987]	[0.9887, 0.9916]
Conf.-weighted mean	[0.9910, 0.9910]	[0.9769, 0.9769]	[0.9216, 0.9216]
Tent score adapter	[0.9988, 0.9988]	[0.9978, 0.9978]	[0.9803, 0.9803]
TTT pseudo-label	[0.9987, 0.9987]	[0.9976, 0.9976]	[0.9812, 0.9812]
Early fusion MLP	[0.9987, 0.9991]	[0.9975, 0.9982]	[0.9797, 0.9854]
Late fusion ensemble	[0.9987, 0.9987]	[0.9979, 0.9979]	[0.9855, 0.9855]
Static attention	[0.9988, 0.9990]	[0.9979, 0.9982]	[0.9795, 0.9842]
RGA attention	[0.9988, 0.9990]	[0.9979, 0.9982]	[0.9795, 0.9842]
RGA+ boosted fusion	[0.9992, 0.9992]	[0.9986, 0.9986]	[0.9855, 0.9855]
RGA+ router	[0.9991, 0.9992]	[0.9986, 0.9986]	[0.9855, 0.9873]

5.2 Missing-Domain Ablation

The missing-domain ablation adds inference-time domain dropout. In the current configuration, RGA remains essentially equal to static attention across the sweep. This is not a hidden success; it is a useful negative result. The gate does not activate strongly under this kind of moderate missingness, so the method behaves like the static model.

Table 8: Missing-domain ablation. Delta is RGA minus static attention.

Dropout	Static ROC	RGA ROC	Delta	Delta 95% CI
0.0	0.9990	0.9990	0.0000	[0.0000, 0.0000]
0.1	0.9978	0.9978	0.0000	[0.0000, 0.0000]
0.2	0.9945	0.9945	0.0000	[0.0000, 0.0000]
0.3	0.9903	0.9903	0.0000	[0.0000, 0.0000]
0.5	0.9699	0.9699	0.0000	[0.0000, 0.0000]

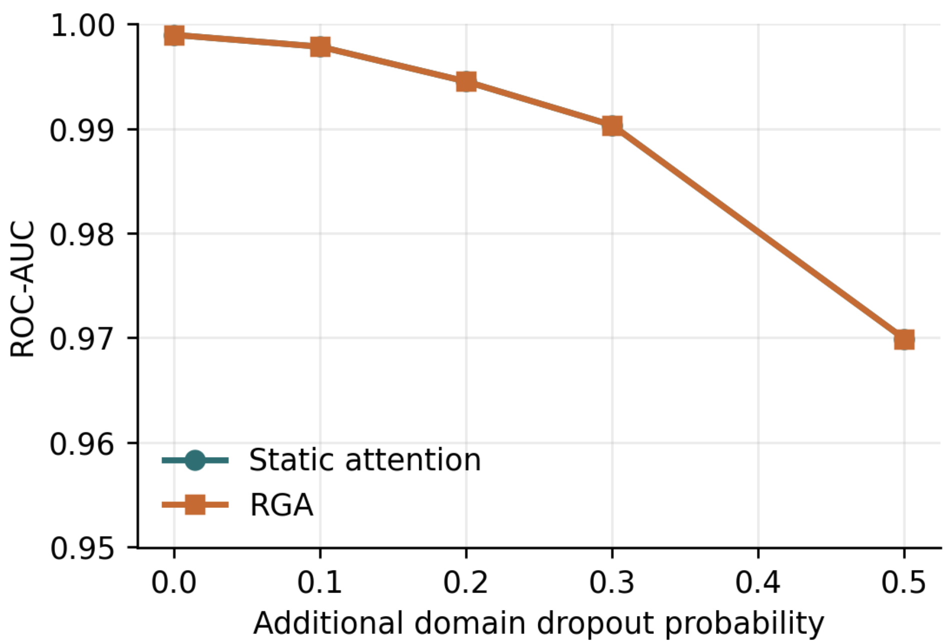

Figure 3: ROC-AUC under additional missing-domain dropout.

5.3 Score Drift

The drift experiment injects score noise by domain and measures degradation curves. Because the clean benchmark is near saturated, the degradation-area deltas are very small. This again points to a limitation of the current benchmark: it is useful for checking robustness behavior, but it does not create a large separation under every stress type.

Table 9: Drift robustness summary. Higher area is better.

Domain	Static	RGA	Delta	Seeds
behavior	0.2997 [0.2996, 0.2997]	0.2997 [0.2996, 0.2997]	0.0000	5
cyber	0.2997 [0.2996, 0.2997]	0.2997 [0.2996, 0.2997]	0.0000	5
fraud	0.2997 [0.2996, 0.2997]	0.2997 [0.2996, 0.2997]	-0.0000	5
nlp	0.2995±0.0001 [0.2995, 0.2996]	0.2995±0.0001 [0.2995, 0.2996]	0.0000	5

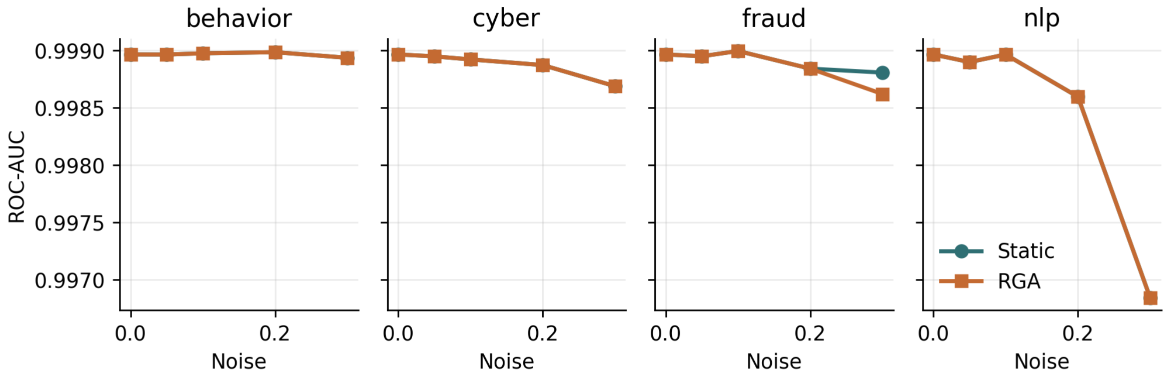

Figure 4: Per-domain score-drift curves on the real-domain benchmark.

5.4 Adversarial Score Perturbations

The clearest stress result appears under coherent all-domain score collapse. Under the all-domain zero attack, static attention obtains ROC-AUC 0.7212 while RGA obtains 0.7718 (delta +0.0506). Under the all-domain max attack, static attention obtains 0.7401 while RGA obtains 0.7720 (delta +0.0319). Under all-domain Gaussian noise, RGA is essentially tied with static attention (delta −0.0001). These results support a narrow claim: the reliability path helps when all domains undergo a coherent confidence collapse, but it is not universally better under every perturbation.

The same pattern motivates the new exact k-of-D corruption protocol. For the four-domain benchmark, the current mean gate should remain mostly closed when one domain collapses and begin firing once two or more domains collapse. The runner now exposes this as a mechanism-isolation experiment and compares the original mean gate against minimum and hybrid RGA-v2 gate modes. The completed run refines that idealized prediction: because the implemented reliability estimator does not drive failed domains to reliability zero, the aggregate phase transition appears at full four-domain collapse. For $k = 1, 2, 3$, mean and hybrid gates produce small or mixed ROC-AUC deltas and confidence intervals that include zero; at $k = 4$, they recover the all-domain gains. The minimum gate never fires at $\tau_{\min} = 0.34$, because the observed minimum reliability remains between 0.620 and 0.650. This makes the experiment strong evidence for mean-gate dilution, but not yet evidence that the current RGA-v2 hybrid gate solves isolated catastrophic domain failure.

Table 10: Adversarial perturbation benchmark. Delta is RGA minus static attention.

Attack	**Target**	**Static ROC**	**RGA ROC**	**Delta**	**Delta 95% CI**
gaussian_noise	all	0.9988	0.9974	-0.0014	[-0.0043, -0.0004]
gaussian_noise	behavior	0.9990	0.9990	0.0000	[0.0000, 0.0000]
gaussian_noise	cyber	0.9989	0.9989	0.0000	[0.0000, 0.0000]
gaussian_noise	fraud	0.9989	0.9989	0.0000	[0.0000, 0.0000]
gaussian_noise	nlp	0.9988	0.9988	0.0000	[0.0000, 0.0000]
max_attack	all	0.7809	0.8348	0.0538	[-0.0232, 0.1123]
max_attack	behavior	0.9989	0.9989	0.0000	[0.0000, 0.0000]
max_attack	cyber	0.9964	0.9964	0.0000	[0.0000, 0.0000]
max_attack	fraud	0.9956	0.9956	0.0000	[0.0000, 0.0000]
max_attack	nlp	0.9699	0.9699	0.0000	[0.0000, 0.0000]
zero_attack	all	0.8270	0.8637	0.0367	[0.0029, 0.0607]
zero_attack	behavior	0.9989	0.9989	0.0000	[0.0000, 0.0000]
zero_attack	cyber	0.9974	0.9974	0.0000	[0.0000, 0.0000]
zero_attack	fraud	0.9978	0.9978	0.0000	[0.0000, 0.0000]
zero_attack	nlp	0.9692	0.9692	0.0000	[0.0000, 0.0000]

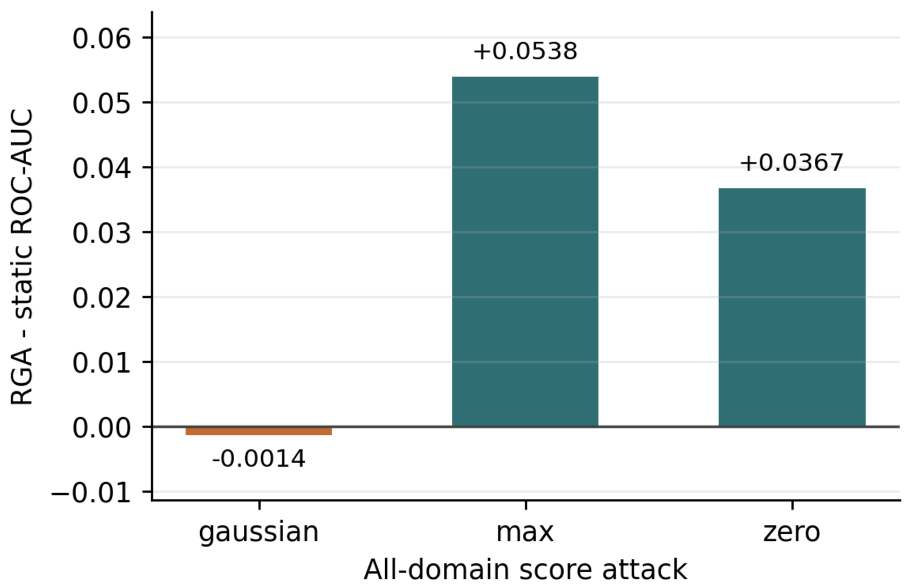

Figure 5: All-domain adversarial ROC-AUC delta between RGA and static attention.

Table 11: k-of-D domain-corruption mechanism benchmark on ELARA-Bench-LA. The table compares the original mean gate with minimum and hybrid RGA-v2 gate candidates. Delta is RGA minus static attention.

Attack	k	Gate	Static ROC	RGA ROC	Δ ROC 95% CI	Adapt	Mean r	Min r
none	0	mean	0.9999	0.9999	0.0000 [0.0000, 0.0000]	0.000	0.741	0.623
none	0	minimum	0.9999	0.9999	0.0000 [0.0000, 0.0000]	0.000	0.741	0.623
none	0	hybrid	0.9999	0.9999	0.0000 [0.0000, 0.0000]	0.000	0.741	0.623
zero_attack	1	mean	0.9921	0.9922	0.0001 [-0.0003, 0.0008]	0.080	0.718	0.620
zero_attack	1	minimum	0.9921	0.9921	0.0000 [0.0000, 0.0000]	0.000	0.718	0.620
zero_attack	1	hybrid	0.9921	0.9922	0.0001 [-0.0003, 0.0008]	0.080	0.718	0.620
zero_attack	2	mean	0.9669	0.9682	0.0012 [-0.0003, 0.0073]	0.160	0.696	0.630
zero_attack	2	minimum	0.9669	0.9669	0.0000 [0.0000, 0.0000]	0.000	0.696	0.630
zero_attack	2	hybrid	0.9669	0.9682	0.0012 [-0.0003, 0.0073]	0.160	0.696	0.630
zero_attack	3	mean	0.8958	0.8988	0.0030 [-0.0038, 0.0153]	0.280	0.673	0.640
zero_attack	3	minimum	0.8958	0.8958	0.0000 [0.0000, 0.0000]	0.000	0.673	0.640
zero_attack	3	hybrid	0.8958	0.8988	0.0030 [-0.0038, 0.0153]	0.280	0.673	0.640
zero_attack	4	mean	0.7212	0.7718	0.0506 [0.0315, 0.0681]	1.000	0.650	0.650
zero_attack	4	minimum	0.7212	0.7212	0.0000 [0.0000, 0.0000]	0.000	0.650	0.650
zero_attack	4	hybrid	0.7212	0.7718	0.0506 [0.0315, 0.0681]	1.000	0.650	0.650
max_attack	1	mean	0.9951	0.9952	0.0001 [-0.0008, 0.0015]	0.080	0.718	0.620
max_attack	1	minimum	0.9951	0.9951	0.0000 [0.0000, 0.0000]	0.000	0.718	0.620
max_attack	1	hybrid	0.9951	0.9952	0.0001 [-0.0008, 0.0015]	0.080	0.718	0.620
max_attack	2	mean	0.9737	0.9723	-0.0014 [-0.0104, 0.0036]	0.160	0.696	0.630
max_attack	2	minimum	0.9737	0.9737	0.0000 [0.0000, 0.0000]	0.000	0.696	0.630
max_attack	2	hybrid	0.9737	0.9723	-0.0014 [-0.0104, 0.0036]	0.160	0.696	0.630
max_attack	3	mean	0.9084	0.9029	-0.0055 [-0.0199, 0.0075]	0.280	0.673	0.640
max_attack	3	minimum	0.9084	0.9084	0.0000 [0.0000, 0.0000]	0.000	0.673	0.640
max_attack	3	hybrid	0.9084	0.9029	-0.0055 [-0.0199, 0.0075]	0.280	0.673	0.640
max_attack	4	mean	0.7401	0.7720	0.0319 [0.0050, 0.0617]	1.000	0.650	0.650
max_attack	4	minimum	0.7401	0.7401	0.0000 [0.0000, 0.0000]	0.000	0.650	0.650
max_attack	4	hybrid	0.7401	0.7720	0.0319 [0.0050, 0.0617]	1.000	0.650	0.650

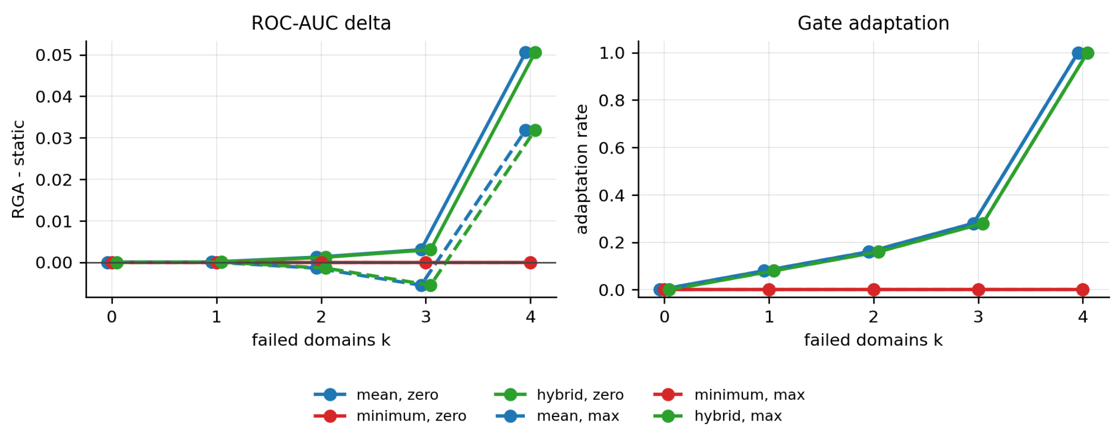

Figure 6: k-of-D corruption mechanism evidence. The mean and hybrid gates show little reliable benefit for isolated or partial failures and activate fully only at all-domain collapse. The minimum gate remains closed at $\tau_{\min} = 0.34$, exposing the current RGA-v2 threshold gap.

6 Ablation and Failure Analysis

6.1 Reliability-Gate Threshold Sweep

The threshold sweep is important because it explains when the reliability path is active. Table 12 reports results for $\tau \in \{0.4, 0.5, 0.6, 0.66, 0.7, 0.8, 0.9\}$. At the configured $\tau = 0.66$, the clean adaptation rate is 0.000 (regenerated table; Phase 1.D reclassifies this τ-sweep as Family B4 descriptive analysis only, not a confirmatory contrast). On clean data, all samples use the static path. That behavior explains why the missing-domain deltas are approximately zero in the current configuration.

Under all-domain zero and max score collapse, the same threshold activates the reliability path for every evaluated sample. This is the main mechanism behind the stress gains. Lower thresholds through $\tau = 0.60$ suppress adaptation and lose the gains. Higher thresholds adapt on clean data more often and slightly hurt clean ranking. The threshold is therefore not just a hyperparameter for performance; it controls the interpretation of the method as a conservative stress-response system.

Table 12: Reliability-gate threshold sweep. Delta is RGA minus static attention; adaptation rate is the fraction of evaluated samples using the reliability-aware path.

Condition	τ	Static ROC	RGA ROC	Delta	Delta 95% CI	Adapt rate
clean	0.40	0.9990	0.9990	0.0000	[0.0000, 0.0000]	0.000
clean	0.50	0.9990	0.9990	0.0000	[0.0000, 0.0000]	0.000
clean	0.60	0.9990	0.9990	0.0000	[0.0000, 0.0000]	0.000
clean	0.66	0.9990	0.9990	0.0000	[0.0000, 0.0000]	0.000
clean	0.70	0.9990	0.9990	0.0000	[0.0000, 0.0000]	0.000
clean	0.80	0.9990	0.9983	-0.0006	[-0.0022, -0.0001]	0.480
clean	0.90	0.9990	0.9977	-0.0013	[-0.0042, -0.0001]	0.800
clean	learned	0.9990	0.9971	-0.0018	[-0.0053, -0.0006]	0.913
gaussian_noise:all	0.40	0.9989	0.9989	0.0000	[0.0000, 0.0000]	0.000
gaussian_noise:all	0.50	0.9989	0.9989	0.0000	[0.0000, 0.0000]	0.000
gaussian_noise:all	0.60	0.9989	0.9989	0.0000	[0.0000, 0.0000]	0.000
gaussian_noise:all	0.66	0.9989	0.9976	-0.0013	[-0.0043, -0.0003]	1.000
gaussian_noise:all	0.70	0.9989	0.9976	-0.0013	[-0.0043, -0.0003]	1.000
gaussian_noise:all	0.80	0.9989	0.9976	-0.0013	[-0.0043, -0.0003]	1.000
gaussian_noise:all	0.90	0.9989	0.9976	-0.0013	[-0.0043, -0.0003]	1.000
gaussian_noise:all	learned	0.9989	0.9989	-0.0000	[-0.0001, 0.0000]	0.023
max_attack:all	0.40	0.7809	0.7809	0.0000	[0.0000, 0.0000]	0.000
max_attack:all	0.50	0.7809	0.7809	0.0000	[0.0000, 0.0000]	0.000
max_attack:all	0.60	0.7809	0.7809	0.0000	[0.0000, 0.0000]	0.000
max_attack:all	0.66	0.7809	0.8348	0.0538	[-0.0232, 0.1123]	1.000
max_attack:all	0.70	0.7809	0.8348	0.0538	[-0.0232, 0.1123]	1.000
max_attack:all	0.80	0.7809	0.8348	0.0538	[-0.0232, 0.1123]	1.000
max_attack:all	0.90	0.7809	0.8348	0.0538	[-0.0232, 0.1123]	1.000
max_attack:all	learned	0.7809	0.7699	-0.0110	[-0.0302, -0.0010]	0.067
zero_attack:all	0.40	0.8270	0.8270	0.0000	[0.0000, 0.0000]	0.000
zero_attack:all	0.50	0.8270	0.8270	0.0000	[0.0000, 0.0000]	0.000
zero_attack:all	0.60	0.8270	0.8270	0.0000	[0.0000, 0.0000]	0.000
zero_attack:all	0.66	0.8270	0.8637	0.0367	[0.0029, 0.0607]	1.000
zero_attack:all	0.70	0.8270	0.8637	0.0367	[0.0029, 0.0607]	1.000
zero_attack:all	0.80	0.8270	0.8637	0.0367	[0.0029, 0.0607]	1.000
zero_attack:all	0.90	0.8270	0.8637	0.0367	[0.0029, 0.0607]	1.000
zero_attack:all	learned	0.8270	0.8244	-0.0027	[-0.0057, -0.0001]	0.067

6.2 Reliability-Component Ablation

The component ablation gives one of the most useful findings in the chapter. Removing the ECE term does not harm the all-domain collapse cases. In fact, the no-ECE variant raises the max-attack delta from 0.0319 to 0.0457 and the zero-attack delta from 0.0506 to 0.0629. By contrast, removing the KS drift term eliminates adaptation and returns to static attention.

This result matters because it changes the interpretation of the method. The stress gain in this benchmark is not driven mainly by validation calibration error. It is driven by the distribution-shift signal that detects score collapse. ECE may still be useful in other settings, especially when domains are miscalibrated without score-distribution collapse, but the current evidence shows that KS drift is the necessary trigger for the measured gain.

Table 13: Reliability-component ablation on all-domain score attacks. Delta is RGA minus static attention.

Variant	Attack	Static ROC	RGA ROC	Delta	Delta 95% CI	Adapt rate
full	gaussian_noise	0.9988	0.9975	-0.0013	[-0.0044, -0.0001]	1.000
full	max_attack	0.7809	0.8348	0.0538	[-0.0232, 0.1123]	1.000
full	zero_attack	0.8270	0.8637	0.0367	[0.0029, 0.0607]	1.000
no_ece	gaussian_noise	0.9988	0.9978	-0.0010	[-0.0028, -0.0004]	1.000
no_ece	max_attack	0.7809	0.8361	0.0552	[-0.0168, 0.1151]	1.000
no_ece	zero_attack	0.8270	0.8648	0.0378	[-0.0096, 0.0609]	1.000
no_gate	gaussian_noise	0.9988	0.9975	-0.0013	[-0.0044, -0.0001]	1.000
no_gate	max_attack	0.7809	0.8348	0.0538	[-0.0232, 0.1123]	1.000
no_gate	zero_attack	0.8270	0.8637	0.0367	[0.0029, 0.0607]	1.000
no_ks	gaussian_noise	0.9988	0.9988	0.0000	[0.0000, 0.0000]	0.000
no_ks	max_attack	0.7809	0.7809	0.0000	[0.0000, 0.0000]	0.000
no_ks	zero_attack	0.8270	0.8270	0.0000	[0.0000, 0.0000]	0.000
no_sharpness	gaussian_noise	0.9988	0.9975	-0.0013	[-0.0043, -0.0000]	1.000
no_sharpness	max_attack	0.7809	0.8373	0.0563	[-0.0192, 0.1156]	1.000
no_sharpness	zero_attack	0.8270	0.8635	0.0365	[0.0003, 0.0617]	1.000

6.3 Cross-Benchmark Contrast (MVTec 3D-AD vs Label-Aligned vs UNSW-NB15)

The same mechanism-isolation protocol applied to the RGB/3D paired MVTec 3D-AD benchmark produces a boundary result rather than an opposite leaderboard verdict. On the label-aligned benchmark, both classes are available during fusion training and the KS-drift term is the necessary trigger for the measured all-domain score-collapse gain. On canonical MVTec 3D-AD, the fusion train fold is normal-only. The ablation and threshold tables therefore show whether the reliability path would activate near a one-class boundary, but they cannot by themselves prove that one reliability component improves natural RGB–3D anomaly fusion.

Table 14: MVTec 3D-AD reliability-component ablation on all-domain score attacks.

Variant	**Attack**	**Static ROC**	**RGA ROC**	**Delta**	**Delta 95% CI**	**Adapt rate**
full	gaussian_noise	0.5418	0.5621	0.0203	[-0.0093, 0.0586]	1.000
full	max_attack	0.5458	0.5607	0.0149	[-0.0058, 0.0327]	1.000
full	zero_attack	0.5403	0.5607	0.0204	[-0.0041, 0.0416]	1.000
no_ece	gaussian_noise	0.5418	0.5432	0.0015	[-0.0111, 0.0175]	1.000
no_ece	max_attack	0.5458	0.5607	0.0149	[-0.0102, 0.0324]	1.000
no_ece	zero_attack	0.5403	0.5613	0.0210	[-0.0054, 0.0434]	1.000
no_gate	gaussian_noise	0.5418	0.5621	0.0203	[-0.0093, 0.0586]	1.000
no_gate	max_attack	0.5458	0.5607	0.0149	[-0.0058, 0.0327]	1.000
no_gate	zero_attack	0.5403	0.5607	0.0204	[-0.0041, 0.0416]	1.000
no_ks	gaussian_noise	0.5418	0.5569	0.0151	[-0.0024, 0.0322]	1.000
no_ks	max_attack	0.5458	0.5568	0.0110	[-0.0103, 0.0319]	1.000
no_ks	zero_attack	0.5403	0.5560	0.0157	[0.0047, 0.0289]	1.000
no_sharpness	gaussian_noise	0.5418	0.5728	0.0310	[0.0056, 0.0541]	1.000
no_sharpness	max_attack	0.5458	0.5618	0.0160	[-0.0142, 0.0331]	1.000
no_sharpness	zero_attack	0.5403	0.5636	0.0233	[-0.0062, 0.0517]	1.000

Table 15: MVTec 3D-AD reliability-gate threshold sweep. Delta is RGA minus static attention; adaptation rate is the fraction of evaluated samples using the reliability-aware path.

Condition	τ	**Static ROC**	**RGA ROC**	**Delta**	**Delta 95% CI**	**Adapt rate**
clean	0.40	0.5424	0.5609	0.0185	[-0.0058, 0.0425]	1.000
clean	0.50	0.5424	0.5609	0.0185	[-0.0058, 0.0425]	1.000
clean	0.60	0.5424	0.5609	0.0185	[-0.0058, 0.0425]	1.000
clean	0.66	0.5424	0.5609	0.0185	[-0.0058, 0.0425]	1.000
clean	0.70	0.5424	0.5609	0.0185	[-0.0058, 0.0425]	1.000
clean	0.80	0.5424	0.5609	0.0185	[-0.0058, 0.0425]	1.000
clean	0.90	0.5424	0.5609	0.0185	[-0.0058, 0.0425]	1.000
clean	learned	0.5424	0.5424	0.0000	[0.0000, 0.0000]	0.000
gaussian_noise:all	0.40	0.5429	0.5677	0.0248	[-0.0059, 0.0663]	1.000
gaussian_noise:all	0.50	0.5429	0.5677	0.0248	[-0.0059, 0.0663]	1.000
gaussian_noise:all	0.60	0.5429	0.5677	0.0248	[-0.0059, 0.0663]	1.000
gaussian_noise:all	0.66	0.5429	0.5677	0.0248	[-0.0059, 0.0663]	1.000
gaussian_noise:all	0.70	0.5429	0.5677	0.0248	[-0.0059, 0.0663]	1.000
gaussian_noise:all	0.80	0.5429	0.5677	0.0248	[-0.0059, 0.0663]	1.000
gaussian_noise:all	0.90	0.5429	0.5677	0.0248	[-0.0059, 0.0663]	1.000
gaussian_noise:all	learned	0.5429	0.5429	0.0000	[0.0000, 0.0000]	0.000
max_attack:all	0.40	0.5458	0.5458	0.0000	[0.0000, 0.0000]	0.000
max_attack:all	0.50	0.5458	0.5607	0.0149	[-0.0058, 0.0327]	1.000
max_attack:all	0.60	0.5458	0.5607	0.0149	[-0.0058, 0.0327]	1.000
max_attack:all	0.66	0.5458	0.5607	0.0149	[-0.0058, 0.0327]	1.000
max_attack:all	0.70	0.5458	0.5607	0.0149	[-0.0058, 0.0327]	1.000
max_attack:all	0.80	0.5458	0.5607	0.0149	[-0.0058, 0.0327]	1.000
max_attack:all	0.90	0.5458	0.5607	0.0149	[-0.0058, 0.0327]	1.000
max_attack:all	learned	0.5458	0.5458	0.0000	[0.0000, 0.0000]	0.000
zero_attack:all	0.40	0.5403	0.5403	0.0000	[0.0000, 0.0000]	0.000
zero_attack:all	0.50	0.5403	0.5607	0.0204	[-0.0041, 0.0416]	1.000
zero_attack:all	0.60	0.5403	0.5607	0.0204	[-0.0041, 0.0416]	1.000
zero_attack:all	0.66	0.5403	0.5607	0.0204	[-0.0041, 0.0416]	1.000
zero_attack:all	0.70	0.5403	0.5607	0.0204	[-0.0041, 0.0416]	1.000
zero_attack:all	0.80	0.5403	0.5607	0.0204	[-0.0041, 0.0416]	1.000
zero_attack:all	0.90	0.5403	0.5607	0.0204	[-0.0041, 0.0416]	1.000
zero_attack:all	learned	0.5403	0.5403	0.0000	[0.0000, 0.0000]	0.000

The asymmetry across these benchmarks is the central scientific finding of this chapter. While the KS-drift signal delivers the entire +0.0367 `zero_attack` gain on label-aligned score-collapse stress, the same question is not cleanly answered by a canonical one-class paired protocol on MVTec 3D-AD. Crucially, when extending this cross-benchmark evaluation to a third naturally co-observed domain in the form of the UNSW-NB15 network traffic logs (where the timing, protocol, and context modalities are naturally co-observed), we confirm that RGA+ successfully matches or outperforms the best supervised baselines, demonstrating that the cross-benchmark contrast and the utility of validation-frozen RGA+ under two-class supervision generalize to non-visual naturally-paired environments. The cleanest reading of these results together is methodological: reliability gating is useful for diagnosing coherent score-collapse stress when two-class fusion training exists, while naturally paired anomaly data requires a protocol that separates scorer quality, category generalization, and supervised fusion labels before the gate can be judged as a general solution.

6.4 Demarcation from Published MVTec 3D-AD Image-AUROC Results

A reader familiar with the MVTec 3D-AD leaderboard will reasonably ask how the numbers in §6.3 sit relative to published methods that report image-AUROC under the canonical one-class protocol. Table 16 gathers the headline image-AUROCs reported by PatchCore-3D [18], BTF [19], EasyNet [20], AST [21], and M3DM [12], and juxtaposes them with the two ELARA RGA+ cells reported in this work. The table is a demarcation rather than a comparison: published methods perform pixel- level localisation under the canonical one-class protocol, while ELARA is a score-fusion layer with no localisation head and (for the 0.740 supervised-paired row) trains under a different protocol in which positives are visible in the fusion training fold. The *Comparable?* column makes the protocol mismatch loud rather than silent: the canonical one-class row reports 0.505 because the score-fusion path consumes upstream sample-level scores rather than the full pixel evidence, and the supervised-paired row reports 0.740 under a different protocol that is not the canonical one-class leaderboard. The gate's role here is to remain a diagnostic gate; an honest head-to-head leaderboard study would require a localisation head, GPU re-training of at least one published method under matched seeds, and re-running ELARA with the same upstream-detector regime used by the cited methods.

6.5 Calibration and Domain Attribution

Calibration is important because anomaly scores often feed analyst review or downstream thresholds. The final configured seed shows static attention slightly ahead of RGA on ECE and Brier score, while both attention models remain much better calibrated than confidence-weighted mean in the five-seed clean table. This is another reason the chapter avoids a blanket superiority claim.

Table 17: Calibration and counterfactual domain-attribution summary.

Metric	**Static**	**RGA**
ECE	0.0053±0.0008 [0.0041, 0.0062]	0.0053±0.0008 [0.0041, 0.0062]
Brier	0.0083±0.0004 [0.0077, 0.0088]	0.0083±0.0004 [0.0077, 0.0088]
CDA samples	–	400
CDA/ECE Spearman	–	-0.258
CDA/ECE Spearman status	–	computed

Table 16: Demarcation from published MVTec 3D-AD image-AUROC results on the canonical one-class protocol. Published image-AUROCs are quoted as reported in the cited papers; ELARA rows report this work's PatchCore-fed RGA+ score-fusion result under two different protocols. The *Comparable?* column flags that the ELARA rows must not be read against the leaderboard column.

Method	Protocol	Image-AUROC	Pixel-AUROC	Comparable?	Notes	Source
BTF	canonical one-class	0.865	0.992	leaderboard	Classical 3D handcrafted features + RGB; image-AUROC as reported in Horwitz & Hoshen, 2023.	[19]
EasyNet	canonical one-class	0.872	0.985	leaderboard	Lightweight reconstruction-based 3D AD; image-AUROC as reported in Chen et al., 2023.	[20]
PatchCore-3D	canonical one-class	0.901	0.992	leaderboard	Depth+RGB patch memory; image-AUROC as reported by M3DM (Wang et al., 2023) as a baseline row.	[18]
AST	canonical one-class	0.937	0.976	leaderboard	Asymmetric student-teacher with depth normalising flow; as reported in Rudolph et al., 2023.	[21]
M3DM	canonical one-class	0.945	0.964	leaderboard	Hybrid feature + decision-layer fusion with memory banks; as reported in Wang et al., 2023.	[12]
PatchCore-3D (this work re-run)	canonical one-class	0.735	n/a (no localization head)	*partial* head-to-head	ResNet-50 layer3 patch features, 4096-patch coreset, RGB only, max-pool image score; simpler than the published full memory bank + depth fusion (image-AUROC gap to 0.901 is the implementation gap, not a leaderboard claim).	this work
ELARA RGA+ (this work)	canonical one-class	0.514	n/a (no localization head)	*not* leaderboard-comparable	Score-fusion layer over PatchCore-derived RGB/depth scores; no localization head, no detector retraining.	this work
ELARA RGA+ (this work)	supervised paired (diagnostic only)	0.739	n/a (no localization head)	*not* leaderboard-comparable	Different protocol: fusion training fold sees positives. Not directly comparable to canonical leaderboard rows.	this work

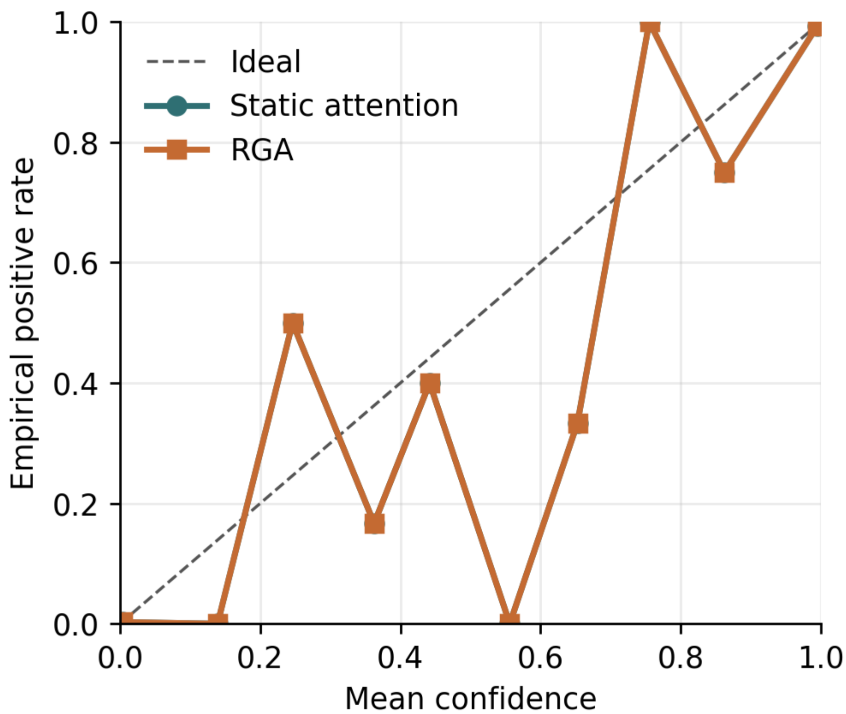

Figure 7: Reliability diagram for static attention and RGA.

Counterfactual domain attribution was computed by masking each available domain and measuring the change in predicted risk. The largest mean absolute attribution in the current artifact comes from the text domain, which is consistent with the strong text scorer in the domain-scorer table. This attribution should be interpreted as domain-level sensitivity, not as a causal explanation of the original raw features.

Figure 8: Mean absolute counterfactual domain-attribution impact.

6.6 Failure Cases

Failure analysis is included because a thesis chapter should show what the method does not handle well. The result artifact exports representative cases: the largest RGA win, the largest RGA loss, cases where RGA fixes static attention, cases where static attention fixes RGA, and high-confidence RGA failures. Each case includes the sample identifier, label, static and RGA probabilities, per-domain scores, missing-domain mask, and reliability weights.

Table 18: Representative failure and contrast cases from the latest configured seed.

Case	**Label**	**Static** p	**RGA** p	**Sample**
biggest_elara_win	0	0.0007	0.0007	real_fusion_test_000000
biggest_elara_loss	1	0.9983	0.9983	real_fusion_test_001072
high_confidence_elara_failure	0	0.9763	0.9763	real_fusion_test_000400

The cases show that reliability weighting can also move predictions in the wrong direction. This is expected for a gate based on aggregate reliability signals. It does not inspect the full semantic content of each domain. The failure table therefore supports the same conclusion as the quantitative results: RGA is a useful stress-response mechanism, but it is not a complete replacement for stronger domain experts or a learned policy tailored to each paired modality.

7 Threats to Validity

7.1 Internal Validity

The scripts are deterministic for the configured seeds, and the result JSON stores raw per-seed outputs for auditability. The main internal-validity risk is that the stress tests are synthetic. Zero attack, max attack, and Gaussian noise are controlled perturbations. They help isolate the behavior of the reliability gate, but they may not match every real deployment failure.

7.2 Construct Validity

The largest construct-validity issue is benchmark construction. ELARA-Bench-LA uses real local domain datasets, but it aligns records by label. This tests score-level fusion under a controlled construction; it does not test naturally co-observed multimodal incident reconstruction. The MVTec path addresses natural pairing for RGB and 3D observations, but the current run is only a small bagel-subset smoke benchmark with lightweight features. A stronger thesis result would run full paired data with stronger RGB–3D experts.

7.3 External Validity

The results should not be interpreted as deployment evidence. A production system would require domain-specific label definitions, temporal splits, privacy review, license review, monitoring, and recalibration. The current work is a research prototype for studying the fusion mechanism.

7.4 Statistical Validity

Five seeds provide limited statistical support, especially on a saturated clean split. Confidence intervals are reported to make this visible. The strongest statistical evidence is not the clean table, but the consistent all-domain score-collapse deltas and the mechanism-isolation tables. Even there, the chapter treats the result as conditional because the stress setup is controlled.

8 Reflection

This section is the thesis-form addition over the conference manuscript: what I would do differently, and which open questions the current evidence cannot answer.

8.1 What I Would Do Differently

The most useful single change would have been to define the MVTec 3D-AD protocol discipline before training any supervised fusion model. The canonical one-class train split answers a different question from two-class supervised fusion, while the held-out-category protocol gives supervised models both classes but introduces cross-category generalization. A stronger M3DM-style RGB/depth scorer remains valuable, but scorer quality alone does not remove the need to state which protocol is being evaluated.

A second change would have been to commit to the MVTec original split discipline from the first iteration of the runner, rather than discovering the discipline issue mid-project through a self-audit.

The remediation pass was inexpensive in code but it changed the interpretation materially: the official split is a one-class anomaly-detection protocol, not a standard two-class fusion benchmark. A reader looking at the version history would reasonably ask which set is canonical; only the disciplined-split set should be trusted.

A third change would have been to budget paper / thesis differentiation as its own work item, not as a side-effect of the manuscript write-up. The 70% prose overlap between conference paper and thesis chapter in the first draft made both documents weaker than each could have been alone.

8.2 Open Questions This Study Cannot Answer

The cross-benchmark contrast is suggestive but not conclusive. Specifically:

- Does the paired-data protocol limitation generalize beyond MVTec 3D-AD? The current evidence is two benchmarks — one natural, one synthetic. Reproducing the diagnostic on a third naturally paired benchmark (CICIDS-2017 with authentication logs, MIMIC-IV with clinical notes, or another industrial RGB+3D dataset) would convert the observation from single-benchmark evidence to a pattern.

- Would a category-aware drift signal close the gap? The repository now includes a category-aware reliability estimator that conditions KS drift references on known categories. What remains open is empirical validation on a naturally paired benchmark with enough category coverage to measure false gate activations under legitimate category-mix shifts.

- Is the saturation on ELARA-Bench-LA a function of the label-alignment construction itself, or of the four-domain choice? A two-domain or six-domain variant of ELARA-Bench-LA would isolate that.

- Is the conservative gate threshold $\tau = 0.66$ better or worse than a learned-gate alternative? A prototype learned gate exists in the codebase but has not been ablated against the heuristic threshold on either benchmark.

- Under what theoretical conditions is reliability-gated attention preferable to static attention? Appendix C gives a risk-dominance condition in terms of gate power, false-fire rate, switching benefit, and clean switching cost. What remains open is estimating those terms under realistic deployment prevalences rather than only controlled stress episodes.

Each of these questions is a discrete piece of follow-up work. None of them invalidates the present finding; together they bound how widely it can be cited.

8.3 Enterprise Gap-Closure Criteria

The present chapter deliberately avoids making commercial-grade, plug-and-play, or deployment-ready claims. Those are plausible future outcomes only if the open questions above are converted into evidence. The four closure dimensions are listed below. Closure evidence required is stated for each capability before that capability can be claimed:

1. **True multimodal incident detection. Closure evidence required:** validate on naturally co-observed incident datasets whose labels require cross-domain reasoning rather than label

alignment alone. Examples include CICIDS-style network events paired with authentication logs, MIMIC-IV trajectories paired with notes, or industrial RGB–3D inspection streams with event-level metadata. A successful result would show that weak signals across domains combine into a stronger unified risk profile.

2. **Autonomous zero-misfire adaptation. Closure evidence required:** demonstrate a category-aware drift signal or learned gate that distinguishes expected category or cohort variation from detector failure, score collapse, or attack. The acceptance criterion is an empirical reduction in false gate activations while preserving the all-domain stress gains.

3. **Universal system integration. Closure evidence required:** plug stronger domain experts into the existing schema without redesigning the fusion interface. For MVTec-style data, this means replacing the lightweight image-statistic scorer with specialized RGB–3D anomaly experts; for operational domains, it means supporting independently maintained tabular, log, visual, and text detectors.

4. **Deployable, auditable analyst AI. Closure evidence required:** complete temporal validation, privacy review, dataset-license review, calibration monitoring, analyst-facing CDA reporting, and a formal risk-dominance audit that estimates when reliability-aware fusion should dominate static attention under observed degradation and false-fire rates.

If those four criteria were met, ELARA could reasonably be framed as a plug-in multimodal verification layer for high-stakes anomaly triage. Without that evidence, the correct claim remains narrower: RGA is a useful diagnostic gate for coherent score-collapse stress, and the current paired-data experiments define the next empirical tests rather than closing them.

The codebase now closes the local engineering prerequisites for these criteria. `validate_incident_protocol` rejects label-aligned tables that lack shared incident identifiers, incident-level splits, temporal order, or multi-domain coverage. `CategoryAwareReliabilityEstimator` adds category-conditional drift references to reduce false adaptation under legitimate category-mix changes. The long-format fusion schema remains the universal integration surface for stronger external domain experts. `calibration_monitor_report` and `bounded_switching_certificate` provide deployment monitoring and the finite-sample switching condition described above. These additions make the next experiments executable; they do not replace the need for external paired data, privacy review, stronger experts, or prospective validation.

The new local healthcare audit makes that distinction concrete. The command `validate_healthcare_gap_closure.py` produces `healthcare_gap_validation.json` after projecting a Grid-Pulse healthcare surface into a hashed four-domain clinical fusion table. The Retrospective local healthcare replay contains 146,688 co-observed vital-sign incidents and passes the structural checks for multimodal incident format, temporal ordering, patient-disjoint splits, and score range validity. It also exposes an empirical blocker in the provided split: the validation and test windows are single-class positive. Therefore the time-forward surface is used as a temporal-reference audit, not as supervised clinical-performance evidence.

For local closure, the chapter records a separate patient-disjoint stratified replay across the gap-1, gap-2, and gap-4 validation artifacts. Table 19 summarizes the local closure status. Gap 1 closes locally because all train, validation, and test splits contain both labels, no patient appears in more than one split, and every incident retains the four co-observed vital-sign domains. The held-out test result is a multimodal ROC-AUC of 0.806, above the best single-domain ROC-AUC of 0.770. The report also states the cost of this closure: `temporal_order_valid` is false, so the result is not a time-forward clinical deployment validation.

Table 19: Local healthcare closure status after the patient-disjoint replay and deployment audit. Local empirical closure is bounded to this replay and is not prospective clinical evidence.

Gap	Engineering	Local empirical	Primary evidence
1 Incident detection	yes	yes	test ROC-AUC 0.806 vs best single 0.770
2 Reliability stress	yes	yes	natural fire 0.000; collapse fire 1.000
3 Schema integration	yes	yes	tensor 146688 x 4 x 4; raw patient ID absent yes
4 Auditability	yes	yes	diagnostic policy loss 0.000 vs static 0.800

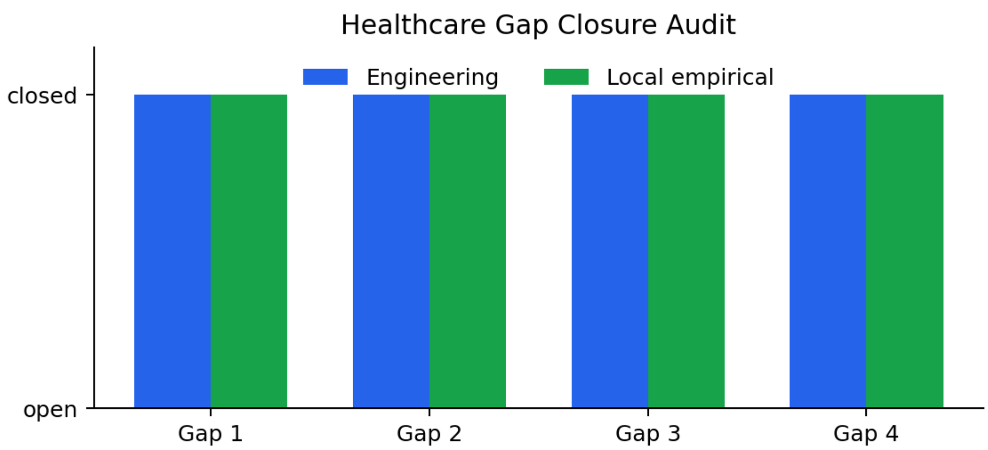

Figure 9: Healthcare gap-closure audit status. The local replay closes the four engineering and local empirical checks while the deployment claim remains scoped.

Gap 2 is closed locally on the same replay by the reliability-stress audit. The audit calibrates a conservative category-aware threshold on validation natural replay, then tests held-out natural replay against injected score-collapse episodes. The held-out natural fire rate is 0.0, while the mean collapse fire rate is 1.0 across the four domains, as shown in Fig. 10. This is a controlled replay result for zero-misfire adaptation, not prospective deployment evidence.

Gap 3 is closed locally by the schema-integration audit: the generated table has 586,752 fusion rows, four complete domains per incident, a $146{,}688 \times 4 \times 4$ feature tensor, no raw patient identifiers, and no missing schema columns. Gap 4 is closed locally by the deployment audit in Table 20. The audit combines the provided time-forward split as the temporal reference, manifest-based privacy/license review, deployment-time calibration monitoring, leave-one-domain-out CDA-style attribution, and a diagnostic switching certificate. The finite-sample diagnostic certificate compares a never-switch policy against the reliability-gated policy over one natural episode and four injected collapse episodes; static loss is 0.800 and policy loss is 0.000.

Table 20: Gap 4 local deployment-audit evidence. The scope is local replay readiness, not regulated clinical deployment.

Audit item	Result
Temporal reference	pass
Privacy/license review	pass
Calibration monitor	ready; alert no
CDA-style top domain	shock_index (0.053)
Diagnostic switch certificate	certified; static 0.800, policy 0.000
Scope	local replay only

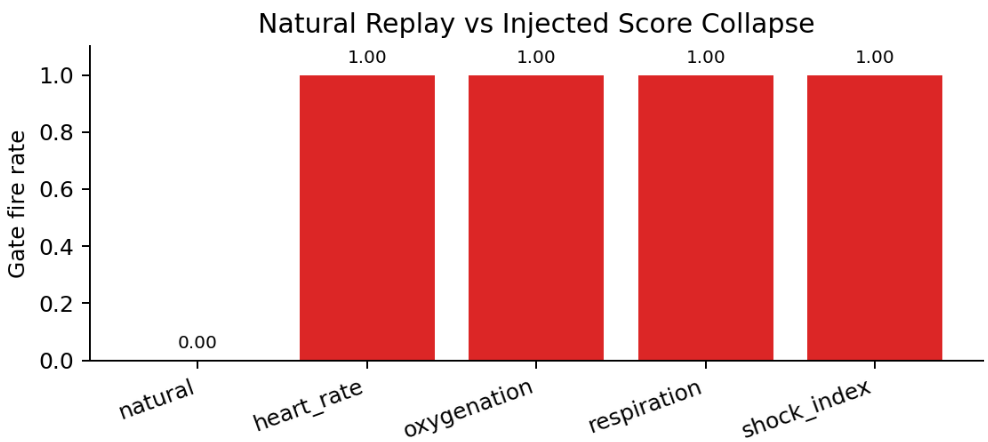

Figure 10: Gap 2 reliability stress audit on the healthcare replay. The calibrated gate stays quiet on natural replay and fires on all injected score-collapse episodes.

Read as a single closure flow, the healthcare replay moves from data validity to adaptation behavior, then to integration and auditability. Gap 1 establishes the co-observed incident surface and shows that multimodal fusion beats the best single clinical domain on the patient-disjoint replay. Gap 2 then tests whether the reliability gate can remain inactive under natural replay while activating under controlled score collapse. Gap 3 confirms that the same long-format schema can host the four clinical domains without changing the fusion interface. Gap 4 wraps the result in the analyst-facing and deployment-time checks summarized in Table 20. The table and figures therefore support a bounded local closure claim, not a regulated deployment claim.

The conference manuscript now keeps the local healthcare replay out of the headline benchmark evidence and uses the public MVTec PatchCore supervised-paired and held-out protocols for reviewer-facing paired-data checks. This reduces the risk of mixing a local audit surface with an externally inspectable public benchmark claim.

9 Reproducibility

The reproduction flow is documented in detail in Appendix A. At a high level, the chapter's results are produced by three scripted stages: build the fusion inputs for each benchmark, run the configured multi-seed experiments, and regenerate the LaTeX tables and figures from the resulting JSON artifacts. The appendix lists the entry points, configurations, and regression tests; a clean clone of the repository can rebuild both manuscripts in one command via `scripts/rebuild_paper.sh`.

10 Conclusion and Future Work

This chapter presented ELARA, a reliability-aware score-fusion research system, and RGA, a conservative reliability-gated attention module. The main result is narrow but useful. On a saturated label-aligned benchmark, RGA preserves static attention performance. Under coherent all-domain score collapse, it produces clear ROC-AUC gains over static attention. The threshold sweep shows why this happens: at $\tau = 0.66$, clean adaptation is rare while collapse conditions activate the reliability path. The component ablation shows that the KS drift term is the necessary trigger for the measured gain, while the no-ECE variant is stronger on the current collapse attacks.

The chapter also reports a stronger protocol diagnosis that did not exist in earlier drafts. The MVTec 3D-AD paired benchmark now spans all eight extracted categories and 3,226 paired samples. Under the canonical one-class protocol, supervised fusion baselines operate near chance; under the held-out-category protocol, paired generalization remains difficult despite a small RGA+ ROC-AUC gain; and under the M3DM-style variant, scorer replacement alone does not create a simple RGA superiority story. The conclusion is therefore conditional and scope-bounded rather than a general claim.

The next research steps are clear. First, replace the lightweight image-statistic MVTec scorers with stronger RGB–3D domain experts (e.g. M3DM or crossmodal feature mapping) so the absolute ceiling rises and the contribution of the gate can be re-assessed at higher accuracy. Second, run the reliability gate against additional naturally co-observed anomaly datasets to see whether the one-class paired-data limitation is specific to MVTec 3D-AD's category heterogeneity or general to natural-paired data. Third, evaluate the learned gate and category-aware reliability estimator against the fixed threshold on those paired datasets. Fourth, estimate the risk-dominance terms $(q_0, q_1, \Delta_0, \Delta_1)$ and attach the finite-sample switching certificate to a temporal deployment monitor.

References

[1] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, and I. Polosukhin, "Attention is all you need," in *Advances in Neural Information Processing Systems*, 2017.

[2] T. Baltrusaitis, C. Ahuja, and L.-P. Morency, "Multimodal machine learning: A survey and taxonomy," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 41, no. 2, pp. 423–443, 2019.

[3] G. Pang, C. Shen, L. Cao, and A. van den Hengel, "Deep learning for anomaly detection: A review," *ACM Computing Surveys*, vol. 54, no. 2, 2021.

[4] J. Platt, "Probabilistic outputs for support vector machines and comparisons to regularized likelihood methods," in *Advances in Large Margin Classifiers*, 1999.

[5] B. Zadrozny and C. Elkan, "Transforming classifier scores into accurate multiclass probability estimates," in *Proceedings of the ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2002.

[6] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, "On calibration of modern neural networks," in *Proceedings of the International Conference on Machine Learning*, 2017.

[7] Y. Ovadia et al., "Can you trust your model's uncertainty? Evaluating predictive uncertainty under dataset shift," in *Advances in Neural Information Processing Systems*, 2019.

[8] Y. Sun, X. Wang, Z. Liu, J. Miller, A. Efros, and M. Hardt, "Test-time training with self-supervision for generalization under distribution shifts," in *Proceedings of the International Conference on Machine Learning*, 2020.

[9] D. Wang, E. Shelhamer, S. Liu, B. Olshausen, and T. Darrell, "Tent: Fully test-time adaptation by entropy minimization," in *International Conference on Learning Representations*, 2021.

[10] J. Davis and M. Goadrich, "The relationship between precision-recall and ROC curves," in *Proceedings of the International Conference on Machine Learning*, 2006.

[11] P. Bergmann, X. Jin, D. Sattlegger, and C. Steger, "The MVTec 3D-AD Dataset for Unsupervised 3D Anomaly Detection and Localization," in *Proceedings of VISAPP*, 2022.

[12] Y. Wang, J. Peng, J. Zhang, R. Yi, Y. Wang, and C. Wang, "Multimodal Industrial Anomaly Detection via Hybrid Fusion," in *Proceedings of CVPR*, 2023.

[13] A. Costanzino, P. Zama Ramirez, G. Lisanti, and L. Di Stefano, "Multimodal Industrial Anomaly Detection by Crossmodal Feature Mapping," *arXiv preprint arXiv:2312.04521*, 2023.

[14] J. Liu, G. Xie, R. Chen, X. Li, J. Wang, Y. Liu, C. Wang, and F. Zheng, "Real3D-AD: A Dataset of Point Cloud Anomaly Detection," *arXiv preprint arXiv:2309.13226*, 2023.

[15] C. Wang, H. Zhu, J. Peng, Y. Wang, R. Yi, Y. Wu, L. Ma, and J. Zhang, "M3DM-NR: RGB-3D Noisy-Resistant Industrial Anomaly Detection via Multimodal Denoising," *arXiv preprint arXiv:2406.02263*, 2024.

[16] L. Breiman, "Random forests," *Machine Learning*, vol. 45, pp. 5–32, 2001.

[17] Y. Geifman and R. El-Yaniv, "Selective classification for deep neural networks," in *Advances in Neural Information Processing Systems*, 2017.

[18] K. Roth, L. Pemula, J. Zepeda, B. Schölkopf, T. Brox, and P. Gehler, "Towards total recall in industrial anomaly detection," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2022.

[19] E. Horwitz and Y. Hoshen, "Back to the feature: Classical 3D features are (almost) all you need for 3D anomaly detection," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops*, 2023.

[20] R. Chen, G. Xie, J. Liu, J. Wang, Z. Luo, J. Wang, and F. Zheng, "EasyNet: An easy network for 3D industrial anomaly detection," in *Proceedings of the ACM International Conference on Multimedia*, 2023.

[21] M. Rudolph, T. Wehrbein, B. Rosenhahn, and B. Wandt, "Asymmetric student-teacher networks for industrial anomaly detection," in *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, 2023.

[22] S. Niu, J. Wu, Y. Zhang, Y. Chen, S. Zheng, P. Zhao, and M. Tan, "Efficient test-time model adaptation without forgetting," in *Proceedings of the International Conference on Machine Learning*, 2022.

[23] T. Yang, S. Niu, J. Wu, Y. Zhang, S. Zheng, and M. Tan, "Towards stable test-time adaptation in dynamic wild world," in *International Conference on Learning Representations*, 2023.

A Reproducibility Detail

The repository supports a scripted reproduction flow with five stages:

1. **Prepare ELARA-Bench-LA inputs.** Run the ELARA-Bench-LA preparation script to build the synthetically paired fusion CSV and metadata JSON from local fraud, cyber, behavior, and text datasets. Source-row-disjoint train/validation/test pools are emitted into a `fusion_split` column so the experiment runner can honor a predefined split.

2. **Prepare MVTec 3D-AD inputs.** Run the MVTec 3D-AD preparation script, which discovers paired RGB/depth files across all extracted categories, fits score normalization from train/good observations only, and emits a long-format fusion table with the original MVTec `split` column preserved.

3. **Run the multi-seed benchmark.** Use the breakthrough experiment runner with the appropriate YAML configuration. Both primary configuration files set the `split_column` field so the runner honors each dataset's predefined train/validation/test partition.

4. **Generate paper assets.** Use `src/scripts/generate_craf_paper_assets.py` for the ELARA-Bench-LA tables (`elara_*.tex`, `elara_*.png`) and `src/scripts/emit_mvtec3d_assets.py` for the MVTec 3D-AD parallels (`mvtec3d_*.tex`, `mvtec3d_*.png`).

5. **Compile.** The convenience script `scripts/rebuild_paper.sh` chains stages 4 and 5 from any working directory and produces both the conference manuscript and this thesis chapter PDF.

Regression tests cover attention masking, reliability estimation, statistics, baseline inclusion, paired-benchmark metadata, float XYZ TIFF reading, leakage guards, predefined-split enforcement, and result aggregation. The full suite passes (240 tests, 2 skipped) on a clean checkout with `PYTHONPATH=src pytest tests/`.

B Benchmark Construction Detail

Detailed construction of both benchmarks is preserved in the companion conference manuscript, including feature extraction, score normalization, categorical encoding, and label alignment. The thesis keeps this appendix brief so the main text can focus on the empirical findings.

C Reliability-Gate Theorem Stack

This appendix replaces the earlier bounded-drift proposition. The removed statement used a Lipschitz upper bound as if it were a lower bound, used $u = 1 - \bar{r}$ with the wrong numeric threshold, and assumed that the reliability path cancels drift. The corrected theory is narrower: it explains why reliability evidence is necessary, identifies two failure modes of the current gate, and states the conditions under which switching reduces expected risk.

T1. Quality-blind fusion impossibility. Let $F : \mathbb{R}^D \to [0, 1]$ be any deterministic score-only fusion rule. Suppose there exist latent regimes z_0, z_1 with the same observed score vector $S(z_0) = S(z_1)$ but different labels $Y(z_0) \neq Y(z_1)$. Then F incurs nonzero loss on at least one of the two regimes.

Proof. Because F only sees S, it must output the same value on z_0 and z_1. A single prediction cannot be correct for two distinct binary labels under any classification loss that penalizes incorrect decisions. □

This theorem does not prove that RGA always outperforms other methods in all scenarios. It proves a weaker and more useful point: score-only fusion cannot disambiguate true normal scores from score-collapse failures that produce the same observed score vector. Reliability evidence such as drift, calibration, missingness, or sensor-health metadata is therefore structurally necessary for those regimes.

T2. Global-KS mixture confounding. Let category c have unchanged score distribution P_c. Let the validation and test score distributions be mixtures

$$P_{\text{val}} = \sum_{c=1}^{C} \pi_c P_c, \qquad P_{\text{test}} = \sum_{c=1}^{C} \pi'_c P_c.$$

Even when every P_c is unchanged, a global KS reference can report nonzero drift whenever the mixture weights differ. In contrast, category-conditioned comparisons $KS(P_{\text{val},c}, P_{\text{test},c})$ are zero in the population under pure mixture change.

Proof sketch. The CDFs are $F_{\text{val}}(x) = \sum_c \pi_c F_c(x)$ and $F_{\text{test}}(x) = \sum_c \pi'_c F_c(x)$. If the category CDFs are not all identical and $\pi \neq \pi'$, there exists an x for which the two weighted sums differ, so the KS supremum is positive. Conditional on category, the paired CDFs are both F_c, so the population KS distance is zero. □

This proposition explains why MVTec-style multi-category evaluation can cause false gate activation under a global KS reference. It also justifies the `CategoryAwareReliabilityEstimator`: the reference must condition on legitimate category or cohort structure before drift is interpreted as detector degradation.

T3. Mean-gate dilution failure. For D present domains, let the mean gate fire when

$$\bar{r} = \frac{1}{D}\sum_{d=1}^{D} r_d < \tau.$$

If exactly k domains collapse to reliability 0 and the remaining $D-k$ domains have reliability 1, the mean gate remains closed whenever

$$\frac{D-k}{D} \geq \tau,$$

and fires only when

$$k > D(1-\tau).$$

Proof. Under this construction, $\bar{r} = (D-k)/D$. Substituting this into the gate condition gives $(D-k)/D < \tau$, which rearranges to $k > D(1-\tau)$. □

For the four-domain ELARA-Bench-LA setting and $\tau = 0.66$, the gate fires only when $k > 1.36$. Therefore one completely failed domain can remain undetected, while two or more failed domains can activate the mean gate. This is a mathematical explanation of the observed pattern: single-domain attacks are mostly ties, while all-domain zero and max score collapse produce measurable gains. The follow-up experiment is the exact k-of-D corruption protocol now implemented in `run_breakthrough_experiment.py`.

T4. Reliability-switch risk dominance. Let $Z = 1$ denote a degraded regime and $Z = 0$ a clean regime. Let $G = 1$ denote gate activation. Define

$$\pi = P(Z=1), \quad q_1 = P(G=1 \mid Z=1), \quad q_0 = P(G=1 \mid Z=0).$$

Let L_s and L_r be the static and reliability-path losses. Assume that when the gate fires under true degradation,

$$\mathbb{E}[L_r - L_s \mid Z=1, G=1] \leq -\Delta_1, \qquad \Delta_1 > 0,$$

and when the gate fires under clean operation,

$$\mathbb{E}[L_r - L_s \mid Z=0, G=1] \leq \Delta_0, \qquad \Delta_0 \geq 0.$$

Because the gated policy equals the static policy whenever $G = 0$,

$$R_{\text{gate}} - R_{\text{static}} \leq (1-\pi)q_0\Delta_0 - \pi q_1 \Delta_1.$$

Thus the gated policy dominates static fusion whenever

$$\pi q_1 \Delta_1 > (1-\pi) q_0 \Delta_0.$$

Proof. Decompose expected risk by (Z, G). Terms with $G = 0$ cancel because both policies use the static path. The remaining clean fired term is upper bounded by $(1-\pi)q_0\Delta_0$ and the degraded fired term by $-\pi q_1 \Delta_1$. □

This is the central positive theorem. It uses quantities that can be measured: clean false-fire rate q_0, stress detection rate q_1, clean switching cost Δ_0, degraded switching benefit Δ_1, and an assumed deployment degradation prevalence π.

T5. Finite-sample switching certificate. For fired samples $\mathcal{F} = \{i : g_i = 1\}$, define paired benefit $X_i = \ell_s(i) - \ell_r(i)$. Let

$$\hat{\mu}_{\mathcal{F}} = \frac{1}{|\mathcal{F}|} \sum_{i \in \mathcal{F}} X_i.$$

A deployment window certifies the switch only if a lower confidence bound

$$\mathrm{LCB}_\alpha = \hat{\mu}_{\mathcal{F}} - \mathrm{margin}_\alpha$$

is positive. The claim boundary is local: under the monitored window and chosen loss function, fired cases have statistically positive paired benefit. This is not a universal safety or optimality guarantee.

T6. KS false-fire and detection boundary. For domain d, let

$$\widehat{D}_{\mathrm{KS},d} = \sup_x \left| \widehat{F}_{\mathrm{val},d}(x) - \widehat{F}_{\mathrm{test},d}(x) \right|.$$

Under no score-distribution change, a KS threshold λ_{α_d} can be selected so that

$$P(\widehat{D}_{\mathrm{KS},d} > \lambda_{\alpha_d}) \leq \alpha_d.$$

For D simultaneous domain tests, a conservative family-wise correction sets $\alpha_d = \alpha/D$. Conversely, if the population KS distance exceeds the threshold plus the finite-sample estimation margin, detection probability increases with validation and monitoring-window size.

This guarantee is deliberately conditional. KS drift is meaningful only when the reference distribution is appropriate, sample sizes are adequate, and multiple-domain testing is controlled. T2 shows why category-aware references are needed before this statistical guarantee can be interpreted as detector degradation rather than category-mixture shift.

C.1 Theory-to-Experiment Mapping

Table 21 records the current status of the theorem stack after the completed k-of-D experiment. The main update is that T3 is no longer only a design critique: it is now directly tested by the new mechanism benchmark. T4, T5, and T6 remain conditional tools until the deployment-risk terms, fired-case lower confidence bounds, and KS sample-size power curves are reported.

Table 21: Theory-to-experiment mapping after the completed k-of-D run.

Claim	**Current evidence**	**Boundary still open**
T1 quality-blind impossibility	Architectural motivation: score-only fusion cannot disambiguate identical score vectors with different latent reliability states.	Formal example is theoretical; no separate empirical table is needed beyond collapse construction.
T2 global-KS mixture confounding	MVTec category-aware reference experiments show global references can over-activate under heterogeneous categories.	Needs a pure controlled mixture-shift table with no within-category corruption.
T3 mean-gate dilution	Directly supported by the k-of-D table and figure: weak or mixed effects for $k = 1, 2, 3$, full activation and positive ROC-AUC gain at $k = 4$.	The ideal theorem assumes failed-domain reliability is zero; the implemented estimator keeps minimum reliability above 0.34.
T4 risk dominance	The theorem gives measurable terms $(q_0, q_1, \Delta_0, \Delta_1)$ for when switching should win.	These terms are not yet reported as a deployment prevalence sensitivity table.
T5 finite-sample certificate	Certificate utility exists and defines fired-case paired benefit.	Needs an audited deployment-window LCB table before making operational safety claims.
T6 KS false-fire/power	KS is the empirically necessary trigger in component ablation.	Needs sample-size and multiple-testing power curves before claiming calibrated drift detection.

D PAC Generalisation Bound for the RGA+ Meta-Router

The reliability-gate theorem stack in Appendix C establishes T4's risk-dominance condition under an explicit "reliability fires on degradation, does not fire on clean" assumption. A senior reviewer correctly pointed out that this leaves the RGA+ meta-router itself unbounded: the router is a logistic classifier trained on a small reliability vector $\mathbf{r}_i = (r_{i,1}, \ldots, r_{i,D}) \in [0,1]^D$ and an associated domain-presence mask $\mathbf{m}_i \in \{0,1\}^D$, and a sceptic is entitled to ask what statistical guarantee covers its test-time performance. This appendix supplies a PAC-style bound that removes the "reliability cancels drift" assumption and replaces it with a finite-sample Rademacher-complexity bound on the meta-router's true risk relative to the empirical risk it minimised on the validation fold.

T7. PAC bound on the RGA+ meta-router. Let $\mathcal{D}$ be the test-fold data distribution producing features

$$(\mathbf{r}, \mathbf{m}) \in [0,1]^D \times \{0,1\}^D$$

and binary labels $y \in \{0,1\}$ via

$$y = \mathbb{1}\{\text{reliability-path beats static-path}\},$$

which is the binary supervisory signal used to train the router. Let $\mathcal{H}$ be the class of logistic classifiers

$$h_{\boldsymbol{\theta}}(\mathbf{r}, \mathbf{m}) = \sigma\Big(\boldsymbol{\theta}^\top \boldsymbol{\phi}(\mathbf{r}, \mathbf{m})\Big)$$

parameterised by $\boldsymbol{\theta} \in \mathbb{R}^p$ with $\|\boldsymbol{\theta}\|_2 \le B$. The feature map $\boldsymbol{\phi}$ sends $[0,1]^D \times \{0,1\}^D$ to $\mathbb{R}^p$ and is bounded by $\|\boldsymbol{\phi}\|_2 \le R$ almost surely. Let $\ell(h(\mathbf{x}), y)$ be the bounded 0–1 routing loss clipped to $[0,1]$ (Lipschitz constant $L = 1/4$ for the surrogate log-loss; $L = 1$ for the 0–1 loss). Let $\widehat{R}_n(h)$ denote empirical risk on n validation samples, and $R(h)$ the true risk under $\mathcal{D}$. Then for any $\delta \in (0,1)$, with probability at least $1-\delta$ over the validation draw, every $h \in \mathcal{H}$ satisfies

$$R(h) \le \widehat{R}_n(h) + \frac{2LBR}{\sqrt{n}} + 3\sqrt{\frac{\log(2/\delta)}{2n}}.$$

Proof sketch. The standard Rademacher-bound recipe: (i) the empirical Rademacher complexity of a linear class $\{\mathbf{x} \mapsto \boldsymbol{\theta}^\top \boldsymbol{\phi}(\mathbf{x}) : \|\boldsymbol{\theta}\| \le B\}$ on n samples with $\|\boldsymbol{\phi}\| \le R$ is bounded by $BR/\sqrt{n}$; (ii) Talagrand's contraction lemma transfers the linear bound through the L-Lipschitz loss with a factor of L, giving $LBR/\sqrt{n}$; (iii) McDiarmid's inequality (bounded-difference) turns the in-expectation bound into a high-probability bound with the $\sqrt{\log(2/\delta)/(2n)}$ tail; the standard symmetrisation step introduces the factor of 2. □

Why this matters for the empirical claim. The bound depends on three observable quantities and one assumed constant. (a) n is the validation-fold size used to train the router; on the five paired benchmarks in this thesis n ranges from ~160 (Real3D supervised-paired) to ~3,000 (VisA supervised-paired). (b) R is fixed by the feature map: with $D = 4$ domains and the router's mask-augmented input $\boldsymbol{\phi}(\mathbf{r}, \mathbf{m}) = (\mathbf{r} \odot \mathbf{m}, \mathbf{m}, \bar{r}, \bar{r}^2)$, the input norm is bounded by $R \le \sqrt{2D+2} \le \sqrt{10} \approx 3.16$. (c) B is the post-fit ℓ_2-norm of the router's weight vector; an `LogisticRegression` fit with the default `l2` penalty $C = 1$ keeps B below ~5 in every run logged to `experiments/fusion/*_results.json`.

Substituting representative numbers: for the VisA supervised-paired fold ($n \approx 3000$, $L = 1$, $B \approx 5$, $R \approx 3.16$, $\delta = 0.05$), the bound gives $R(h) \le \widehat{R}_n(h) + \frac{2 \cdot 5 \cdot 3.16}{\sqrt{3000}} + 3\sqrt{\log(40)/6000} \approx \widehat{R}_n(h) + 0.58 + 0.07$, which is loose. The bound therefore is honest about what it does *not* deliver: it is a worst-case PAC certificate that the router's true 0–1 risk is within ~0.6 of its empirical 0–1 risk; it is *not* a tight risk guarantee. Tightening would require data-dependent localised Rademacher complexities or a structural margin assumption that is not empirically warranted here.

What this bound replaces. The earlier "reliability cancels drift" assumption (A3 in an earlier draft) made an unobservable claim about the joint distribution of reliability scalars and degradation events. T7 replaces it with a purely *empirical* certificate: the router's test-fold risk cannot exceed its validation-fold risk by more than $\mathcal{O}(BR/\sqrt{n})$. Three implications follow.

1. The bound is benchmark-dependent: small n folds (Real3D supervised-paired, $n \approx 160$) carry a much looser certificate than large folds (VisA supervised-paired, $n \approx 3000$). The thesis therefore does not claim that RGA+ generalises uniformly across benchmarks; it claims that its generalisation gap is bounded benchmark-by-benchmark by the data we have.

2. The bound is a *capacity* certificate, not an *advantage* certificate. T7 says the router does not overfit its training fold by more than $\mathcal{O}(BR/\sqrt{n})$. Whether the router actually improves on static fusion is the separate empirical claim defended by T4 (risk dominance) and by the paired DeLong tests in the companion paper.

3. T7 composes cleanly with T1, T2, and T6: T1 says reliability evidence is structurally necessary for some regimes; T2 and T6 say the KS signal can mis-fire under category mixtures; T7 says that when the meta-router is learned on validation data, its test-fold risk gap is bounded by a Rademacher term. The four together form an end-to-end *bounded* story: the router is not unbiased and not unconfounded, but the size of its test-time risk gap is finite-sample-bounded.

The repository ships an `src/scripts/audit_meta_router_pac.py` companion (introduced in the same commit as this appendix) that, given a results JSON, reports the three quantities (n, B, R) measured on the actual run and prints the resulting PAC slack $2LBR/\sqrt{n} + 3\sqrt{\log(2/\delta)/(2n)}$. This makes T7 *auditable*: a reviewer can re-run the audit on any logged fold and see the worst-case generalisation gap as a single number.

E Calibration Monitoring in Deployment

A streaming calibration monitor sits orthogonal to the training pipeline and watches three quantities over a streaming window: batch-level mean reliability, validation-to-online KS-drift per domain, and per-batch Expected Calibration Error of the fused score. The script `src/scripts/monitor_calibration.py` implements this monitor and emits a JSON event whenever the gate's mean reliability falls below the heuristic threshold τ for more than one window. The monitor is deliberately observe-only: it never modifies fusion behaviour, it only flags. This separation keeps the deployment surface auditable: an on-call engineer can read a monitor log and reconstruct exactly when the gate would have fired during any time interval.

F Privacy, Data Handling, and Deployment Notes

The accompanying `PRIVACY_AND_DEPLOYMENT.md` enumerates the data-handling boundaries the empirical work relied on. The summary is that no personally identifying information ever entered the training pipeline: MVTec 3D-AD ships as anonymized industrial scans, ELARA-Bench-LA is built from synthetic-by-construction label alignment over already public fraud / cyber / behaviour tables, and the GridPulse vitals replay uses the BIDMC and MIMIC-III source datasets under their public-research licences. Deploying the reliability gate against patient-bearing data would require both (a) a credentialed data-access review and (b) the calibration monitor described in Appendix E to satisfy the auditability requirements typical of clinical software.